Z-TRANSFORM PROPERTIES AND PAIRS (WITH ROC)

Properties (ROC of X(z) is R: $r_1 < |z| < r_2$)

Property	Time domain	z-domain	ROC
Linearity	$a_1 x_1(n) + a_2 x_2(n)$	$a_1 X_1(z) + a_2 X_2(z)$	at least $R_1 \cap R_2$
Shift (delay k)	$x(n-k)$	$z^{-k}\,X(z)$	R (0 or ∞ may be added or lost)
Scaling	$a^n\,x(n)$	$X(a^{-1}z)$	$\lvert a\rvert r_1 < \lvert z\rvert < \lvert a\rvert r_2$
Time reversal	$x(-n)$	$X(z^{-1})$	$1/r_2 < \lvert z\rvert < 1/r_1$
Differentiation	$n\cdot x(n)$	$-z\cdot$dX(z)/dz	R
Convolution	$x_1(n) * x_2(n)$	$X_1(z)\cdot X_2(z)$	at least $R_1 \cap R_2$ (bigger if a pole cancels)
Correlation	$r_{x_1x_2}(l) = x_1(l) * x_2(-l)$	$X_1(z)\cdot X_2(z^{-1})$	at least $R_1 \cap (1/R_2)$
Multiplication (windowing)	$x_1(n)\cdot x_2(n)$	$(1/2\pi\mathrm{j})\oint X_1(v)\,X_2(z/v)\,v^{-1}\mathrm{dv}$	see text (convolution of transforms)
Initial value (causal x)	$x(0)$	$\lim_{z\to\infty} X(z)$	—

Common transform pairs

x(n)	X(z)	ROC
$\delta(n)$	1	all z
$u(n)$	$1/(1-z^{-1})$	$\lvert z\rvert > 1$
$a^n u(n)$	$1/(1-az^{-1}) = z/(z-a)$	$\lvert z\rvert > \lvert a\rvert$
$-a^n u(-n-1)$	$1/(1-az^{-1})$	$\lvert z\rvert < \lvert a\rvert$
$n\cdot u(n)$	$z^{-1}/(1-z^{-1})^2$	$\lvert z\rvert > 1$
$n\cdot a^n u(n)$	$az^{-1}/(1-az^{-1})^2$	$\lvert z\rvert > \lvert a\rvert$
$-n\cdot a^n u(-n-1)$	$az^{-1}/(1-az^{-1})^2$	$\lvert z\rvert < \lvert a\rvert$
$\sin(\omega_0 n)\,u(n)$	$\sin\omega_0\, z^{-1}/(1-2\cos\omega_0\, z^{-1}+z^{-2})$	$\lvert z\rvert > 1$
$\cos(\omega_0 n)\,u(n)$	$(1-\cos\omega_0\, z^{-1})/(1-2\cos\omega_0\, z^{-1}+z^{-2})$	$\lvert z\rvert > 1$
$a^n \sin(\omega_0 n)\,u(n)$	$a\sin\omega_0\, z^{-1}/(1-2a\cos\omega_0\, z^{-1}+a^2z^{-2})$	$\lvert z\rvert > \lvert a\rvert$
$a^n \cos(\omega_0 n)\,u(n)$	$(1-a\cos\omega_0\, z^{-1})/(1-2a\cos\omega_0\, z^{-1}+a^2z^{-2})$	$\lvert z\rvert > \lvert a\rvert$
finite sequence ($n_1 \dots n_2$)	$\sum x(n)\,z^{-n}$	all z except 0 ($n>0$ terms) and/or ∞ ($n<0$ terms)

QUICK RULES (TRAPS AND SHORTCUTS)

Sampling and the ADC

1. $f' = F/F_s$. Subtract the nearest whole number so $f' \in [-½, ½]$; alias $F' = |f'|\cdot F_s$. **sin flips sign if $f' < 0$**, cos does not. Tones with the same f' add ($3 + 10 = 13$).
2. D/A output: $F = f'\cdot F_s$ (only frequencies inside $\pm F_s/2$ come out). Out-of-band noise folds into the band, so filter **before** the ADC: pass edge B, stop edge $\boldsymbol{F_s - B}$, transition band = $F_s - 2B$ (≥ 0, so $F_s > 2B$ in practice).
3. Quantizer: $\Delta = R/(L-1)$ (from $L = R/\Delta + 1$; the +1 is *outside* the fraction); $L = 2^b$, $b = \lceil \log_2 L \rceil$. Quantize x: $k = \mathrm{round}((x - x_{\min})/\Delta)$, $x_q = x_{\min} + k\Delta$, code = k in binary, error ≤ $\Delta/2$. Sine range = 2A, $P = A^2/2$.

8 bits = 1 byte (kbit/s rate vs kB storage).

Shifts, folds, impulses, windows

1. Every operation **replaces every n** in the current formula: delay $k \to n-k$ (slides right); advance → $n+k$; fold → $-n$. Apply in the order stated: fold-then-delay-4 = $x(-(n-4)) = x(4-n)$; delay-then-fold = $x(-n-4)$ (that is the first result moved 8 left).
2. $\delta(n-k)$ sits at $n=k$. $u(n) = \delta(n) + \delta(n-1) + \delta(n-2) + \dots$ ($n = 0, 1, 2, \dots$). $u(n+3) - u(n-4) = 1$ for $-3 \le n \le 3$ (7 samples). Fold/delay on a formula, e.g. $1 + n/3 \to 1 + (n-4)/3$.
3. Linearity zero-test only **rules out**: $y = 2x + 3$ and cos x fail; $y = x^2$ passes it but is still non-linear (T[2x] ≠ 2 T[x]). Delay = $x(n-k)$; a pure delay is time-invariant, $n\cdot x(n)$ and $x(2n)$ are not.

Convolution and impulse response

1. $y(n) = \sum x(k)\,h(n-k)$: for each n use every pair whose indices **add up to n**. Length $L_x + L_h - 1$, start = sum of the starts, check $\sum y = \sum x \cdot \sum h$. Equals polynomial multiplication in z^{-1}.
2. Finite geometric sum $\sum_{k=0}^{n} r^k = (r^{n+1}-1)/(r-1)$. Example: $(½)^n * (½)^n = (¼)^n(2^{n+1}-1)$. $h(n)$ for $x = \delta(n)$: at every step add the **feedback term** (coefficient × previous h) **and** the input terms; after the input ends $h(n) = p\cdot h(n-1)$.
3. Each input impulse launches a scaled, delayed copy of h: $x = -\delta(n+1) + 2\,\delta(n) + 3\,\delta(n-2) \Rightarrow y = -h(n+1) + 2\,h(n) + 3\,h(n-2)$. Exponent algebra: $a^{n+1} = a\cdot a^n$; combine only **like terms** (same base): $0.5\cdot 0.5^n - 0.5^n = -0.5\cdot 0.5^n$, but $0.5^n - 1^n = 0.5^n - 1$.

LCCDE, ZIR/ZSR and realizations

1. Solve for $y(n)$ to recurse: $y(n) = -\sum a_k y(n-k) + \sum b_m x(n-m)$. Normalize so $y(n)$ has coefficient 1; **a_k = coefficient next to $y(n-k)$ on the left**; drawn feedback gain = **$-a_k$**. Fill a table $n = 0, 1, 2$ (needs $y(-1), \dots$).
2. Homogeneous: λ^n, root of $\lambda^N + a_1\lambda^{N-1} + \dots + a_N = 0$. $y + 0.5\,y(n-1) = x$ has $\lambda =$ **−0.5** (sign!). ZIR: $x = 0$ with the given $y(-1)$; ZSR: $y(-1) = 0$; total = ZI + ZS. Fit C with $y(0)$ from the recursion. Transient = decaying part, steady state = forced part ($H(1)$ × input).
3. DF-I uses $M + N$ delays, DF-II uses $\max(M, N)$. Missing term ⇒ coefficient 0 and no tap.

Z-transform, poles and stability

1. Replace n by $(n-k)$ in time ⇒ z^{-k} in z. Multiply by z^N to get positive powers; the list $[1, a_1, a_2, a_3]$ becomes $\boldsymbol{z^3 + a_1 z^2 + a_2 z + a_3}$ (same order, a_3 is the constant). **$w = z^{-1}$** trick to factor: (1 − ½w)(1 − ¼w) etc.; try ±1, ±2, ±½ as roots, divide out.
2. Calculator polynomial mode: enter a, b, c, d for ax³ + bx² + cx + d, use 0 for a missing power. Complex poles come in conjugate pairs (same magnitude): **magnitude** decides growth, angle only sets frequency.
3. Causal ⇒ ROC outside the outermost pole; stable ⇔ unit circle in ROC ⇒ all $|p| < 1$; pole on the circle = marginal, not BIBO stable. **Quick tests** on monic $f(z)$: $f(1) < 0$ ⇒ real root > 1 (unstable); $(-1)^N f(-1) < 0$ ⇒ root < −1; $|a_N| \ge 1$ ⇒ unstable; $\sum|a_k| < 1$ ⇒ stable. If these pass, compute the roots; they cannot prove stability alone.

PART A · DERIVATIONS (HOW TO ARRIVE AT THE EQUATION)

LECTURE 1 · SAMPLING & QUANTIZATION

Sampling relation and aliasing

1. $x(n) = x_a(\mathrm{n}T) = A\cos(2\pi F\cdot \mathrm{n}T + \varphi) = A\cos(2\pi(F/F_s)n + \varphi)$, so **$f = F/F_s$**.
2. For integer k and integer n: $\cos(2\pi(f+k)n + \varphi) = \cos(2\pi \mathrm{f} n + \varphi)$. So f and $f + k$ give the **same samples**; only $-½ \le f \le ½$ is distinct.
3. Alias: $f' = f - k$ (k = nearest integer), **$F' = F - \mathrm{k}F_s$**. To avoid aliasing need $|F| < F_s/2$: **$F_s \ge 2F_{\max}$**.

SQNR = 6.02b + 1.76 dB

1. Sine of amplitude A: range 2A, $L = 2^b$ levels ⇒ $\Delta \approx 2A/2^b$. Signal power $P_x = A^2/2$.
2. Rounding error e is uniform on $(-\Delta/2, \Delta/2)$: $P_q = (1/\Delta)\int \mathrm{e}^2\,\mathrm{de} = \boldsymbol{\Delta^2/12} = A^2/(3\cdot 2^{2b})$.
3. SQNR = $P_x/P_q = (3/2)\cdot 2^{2b}$. In dB: $10\log_{10}1.5 + 20\,b\cdot\log_{10}2 =$ **$1.76 + 6.02\,b$**.

LECTURE 2 · DT SYSTEMS

The convolution sum

1. Any signal is a sum of shifted impulses: $x(n) = \sum_k x(k)\,\delta(n-k)$.
2. Time invariance: $\delta(n-k) \to h(n-k)$. Linearity: scale and add.
3. **$y(n) = \sum_{k=-\infty}^{\infty} x(k)\,h(n-k) = x(n) * h(n)$**. Substitute $k \to n-k$ to get $h * x$ (commutative). Causal h ($h = 0$ for $n < 0$) and causal x: limits 0 … n.

BIBO stability ⇔ Σ|h(n)| < ∞

1. If $|x| \le M_x$: $|y(n)| \le \sum|h(k)||x(n-k)| \le M_x\sum|h(k)|$. Finite sum ⇒ bounded output (sufficient).
2. Necessary: choose $x(n) = \mathrm{sgn}\,h(-n)$; then $y(0) = \sum|h(k)|$, which is unbounded if the sum is.

LECTURE 3 · LCCDE

Direct method (homogeneous + particular)

1. $y(n) = y_h(n) + y_p(n)$.
2. **Homogeneous** ($x = 0$): try $y_h = \lambda^n$ in $\sum a_k y(n-k) = 0$ ⇒ characteristic polynomial **$\lambda^N + a_1\lambda^{N-1} + \dots + a_N = 0$**. Distinct roots: $\sum C_i\lambda_i^n$; repeated root of multiplicity m: $(C_1 + C_2 n + \dots)\lambda^n$.
3. **Particular:** same form as the input (constant → K; a^n → Ka^n, use nKa^n if a is a root; cos/sin → K_1cos + K_2sin). Substitute to find K.
4. **Constants** C_i from the initial conditions applied to the *total* y (not to y_h). ZIR: $x = 0$ with the given $y(-1)$, $y(-2)$. ZSR: zero initial conditions.

Direct Form II from H(z)

1. $H(z) = B(z)/A(z) = [1/A(z)]\cdot B(z)$: do the all-pole part first. Define $W(z) = X(z)/A(z) \Rightarrow$ **$w(n) = x(n) - a_1 w(n-1) - \dots - a_N w(n-N)$**.
2. Then $Y(z) = B(z)\,W(z) \Rightarrow$ **$y(n) = b_0 w(n) + b_1 w(n-1) + \dots + b_M w(n-M)$**. One shared delay line of $\max(M, N)$ delays.
3. Direct Form I does $B(z)$ first (M delays on x), then $1/A(z)$ (N delays on y): $M + N$ delays.

LECTURE 4 · Z-TRANSFORM

Z{aⁿu(n)} and Z{−aⁿu(−n−1)}, with ROC

1. Causal: $X(z) = \sum_{n=0}^{\infty}(az^{-1})^n =$ **$1/(1 - az^{-1}) = z/(z-a)$** when $|az^{-1}| < 1$, *i.e.* **$|z| > |a|$**.
2. Anti-causal: $-\sum_{n=-\infty}^{-1}(az^{-1})^n = -\sum_{m=1}^{\infty}(z/a)^m = -(z/a)/(1 - z/a) =$ **$1/(1 - az^{-1})$** when **$|z| < |a|$**.
3. Same algebra, different ROC ⇒ different signal. $X(z)$ alone does not determine $x(n)$.

Differentiation: Z{n·x(n)} = −z·dX/dz

1. dX/dz = $\sum(-n)\,x(n)\,z^{-n-1} \Rightarrow -z\cdot$dX/dz = $\sum n\cdot x(n)\,z^{-n}$.
2. Example: $X = 1/(1 - az^{-1}) \Rightarrow$ dX/dz = $-(1 - az^{-1})^{-2}\cdot(az^{-2}) \Rightarrow$ **$Z\{n\cdot a^n u(n)\} = az^{-1}/(1 - az^{-1})^2$**, $|z| > |a|$.

Partial-fraction coefficients and complex poles

1. $X(z) = \sum A_k/(1 - p_k z^{-1})$. Multiply both sides by $(1 - p_k z^{-1})$ and set $z = p_k$: **$A_k = [(1 - p_k z^{-1})X(z)]_{z=p_k}$**.
2. Repeated pole: write $A/(1 - \mathrm{p}z^{-1}) + B/(1 - \mathrm{p}z^{-1})^2$ and use $(1 - \mathrm{p}z^{-1})^{-2} \leftrightarrow (n+1)\,\mathrm{p}^n\,u(n)$ (or the differentiation pair).
3. Complex pair p, p^* ⇒ A, A* (real signal). With A = $|A|e^{j\alpha}$, $p = |p|e^{j\beta}$: **$A\,p^n + A^*(p^*)^n = 2|A||p|^n\cos(\beta n + \alpha)$**.

cos and sin pairs

1. cos $\omega_0 n = ½(e^{j\omega_0 n} + e^{-j\omega_0 n})$: two exponentials a^n with $a = e^{\pm j\omega_0}$.
2. $½[1/(1 - e^{j\omega_0}z^{-1}) + 1/(1 - e^{-j\omega_0}z^{-1})] =$ **$(1 - \cos\omega_0\,z^{-1})/(1 - 2\cos\omega_0\,z^{-1} + z^{-2})$**, $|z| > 1$.
3. sin $\omega_0 n = (1/2\mathrm{j})(e^{j\omega_0 n} - e^{-j\omega_0 n}) \Rightarrow$ **$\sin\omega_0\,z^{-1}/(1 - 2\cos\omega_0\,z^{-1} + z^{-2})$**. For $r^n\cos/r^n\sin$ replace z^{-1} by rz^{-1} (scaling).

Shift, scaling, reversal, convolution

1. **Shift:** $\sum x(n-k)\,z^{-n}$, let $m = n-k$: $z^{-k}\sum x(m)\,z^{-m} = \boldsymbol{z^{-k}\,X(z)}$.
2. **Scaling:** $\sum a^n x(n)\,z^{-n} = \sum x(n)(a^{-1}z)^{-n} = \boldsymbol{X(a^{-1}z)}$; ROC becomes $|a|r_1 < |z| < |a|r_2$.
3. **Reversal:** $\sum x(-n)\,z^{-n}$, let $m = -n$: $\sum x(m)\,z^{m} = \boldsymbol{X(z^{-1})}$; ROC $1/r_2 < |z| < 1/r_1$.
4. **Convolution:** $\sum_n\sum_k x(k)\,h(n-k)\,z^{-n} = \sum x(k)\,z^{-k}\cdot\sum h(m)\,z^{-m}$ ($m = n-k$) = **$\boldsymbol{X(z)\,H(z)}$**.

H(z) from a difference equation, poles and zeros

1. Zero initial conditions, Z of both sides: $Y(z)(1+\Sigma a_k z^{-k}) = X(z)\,\Sigma\, b_m z^{-m} \Rightarrow \boldsymbol{H(z) = B(z)/A(z)}$.
2. Multiply top and bottom by z^N: $H(z) = b_0 z^{N-M}\Pi(z-z_k)/\Pi(z-p_k)$. Zeros z_k: $H=0$. Poles p_k: $H=\infty$ (extra poles/zeros at $z=0$ from $N-M$).
3. Improper ($M \geq N$): divide first, then partial fractions on the remainder. $h(n) = Z^{-1}\{H\}$ for the right ROC.

Stability and causality from the ROC

1. Causal ⇒ right-sided ⇒ ROC is outside the outermost pole ($|z| > r_{\max}$).
2. BIBO stable ⇔ $\Sigma|h| < \infty$ ⇔ $H(e^{j\omega})$ exists ⇔ **unit circle inside the ROC**.
3. Both ⇒ all poles satisfy $\boldsymbol{|p| < 1}$. A pole on the circle is marginal (an oscillator or accumulator), not BIBO stable.

One-sided Z and initial conditions

1. $X^+(z) = \Sigma_{n=0}^{\infty} x(n)\,z^{-n}$. For the delay: $\Sigma_{n=0}^{\infty} x(n-1)\,z^{-n} = x(-1) + z^{-1}X^+(z)$. Twice: $\boldsymbol{Z^+\{x(n-2)\} = z^{-2}X^+(z) + z^{-1}x(-1) + x(-2)}$.
2. Take Z^+ of the LCCDE, collect Y^+: $\boldsymbol{Y^+(z) = P(z)/A(z) + H(z)\,X(z)}$, where P comes only from $y(-1)$, $y(-2)$, ...
3. First term = zero-input response; second = zero-state response. Invert each term separately.

DESIGN DERIVATIONS

First-order smoother from a settling-time spec

1. $y(n) = a{\cdot}y(n-1) + (1-a)\,x(n)$, $0 < a < 1$; DC gain = $(1-a)/(1-a) = 1$.
2. Step response: $y(n) = 1 - a^{n+1}$. To reach 95 % at sample N: $a^{N+1} = 0.05 \Rightarrow \boldsymbol{a = 0.05^{1/(N+1)}}$.
3. Time constant $\tau \approx -1/\ln$ a samples (63 %); 95 % ≈ 3τ.

Moving-average null: zeros at the roots of unity

1. $H(z) = (1/M)\Sigma_{k=0}^{M-1} z^{-k} = (1/M)(1-z^{-M})/(1-z^{-1})$ (geometric sum).
2. Zeros: $z^M = 1$ except $z = 1 \Rightarrow \omega = 2\pi k/M$, $k = 1 \ldots M-1$. To null ω_0 choose $\boldsymbol{M = 2\pi/\omega_0}$ (first zero), which also nulls its multiples.

Digital oscillator (sine and cosine)

1. Want $y(n) = \sin(\omega_0 n)\,u(n)$ for $x = \delta(n)$: $H(z) = Y(z) = \boldsymbol{\sin\,\omega_0\,z^{-1}/(1 - 2\cos\,\omega_0\,z^{-1} + z^{-2})}$, $\omega_0 = 2\pi F/F_s$. Poles at $e^{\pm j\omega_0}$ (on the unit circle).
2. Difference equation: $y(n) = 2\cos\,\omega_0{\cdot}y(n-1) - y(n-2) + \sin\,\omega_0{\cdot}x(n-1)$. Cosine: numerator $1 - \cos\,\omega_0\,z^{-1}$.
3. No impulse? Preload the history: sine $y(-1) = -\sin\,\omega_0$, $y(-2) = -\sin 2\omega_0$; cosine $y(-1) = \cos\,\omega_0$, $y(-2) = \cos 2\omega_0$ (continue the sinusoid backwards).

Inverse system for an echo or ISI

1. Echo: $y(n) = x(n) + \alpha x(n-D) \Rightarrow H(z) = 1 + \alpha z^{-D}$. The equalizer is the inverse: $\boldsymbol{H_i(z) = 1/(1 + \alpha z^{-D})}$.
2. Difference equation: $\boldsymbol{x(n) = y(n) - \alpha x(n-D)}$. $h_i(n) = \Sigma(-\alpha)^k\delta(n - kD)$.
3. Causal and stable only if the poles ($|z|^D = |\alpha|$) lie inside the unit circle: $\boldsymbol{|\alpha| < 1}$.

Notch filter (remove ω₀, leave the rest)

1. Zeros on the circle at $e^{\pm j\omega_0}$ for an exact null: numerator $\boldsymbol{1 - 2\cos\,\omega_0\,z^{-1} + z^{-2}}$.
2. Poles at $r{\cdot}e^{\pm j\omega_0}$, just inside the circle ($r \approx 0.9$–0.99), cancel the zeros away from ω_0: denominator $\boldsymbol{1 - 2r\cos\,\omega_0\,z^{-1} + r^2z^{-2}}$.
3. Bandwidth ≈ $2(1-r)$ rad/sample. DC gain = $H(1) = (2 - 2\cos\,\omega_0)/(1 - 2r\cos\,\omega_0 + r^2)$. Closer $r \to 1$: narrower notch, longer ring-down.

ADC and quantizer sizing

1. Sampling: $F_s \geq 2\,F_{\max}$ (plus a guard band if the anti-alias filter is not ideal).
2. From an SQNR spec: $b \geq (\mathrm{SQNR_{dB}} - 1.76)/6.02$, round up. From a resolution: L = range/Δ + 1 ⇒ $b = \lceil \log_2 L \rceil$.
3. Bit rate = $b{\cdot}F_s$. Memory/delay of D samples costs $D{\cdot}b$ bits; D = delay$\cdot F_s$.

PART B · GRAPHING RULES

Stem plots: shifting, folding, scaling

1. Write the new signal as $x(\text{argument})$. Make a two-row table: n and argument, then read $x(\text{argument})$ at each n.
2. Delay $x(n-k)$: shift right by k. Fold $x(-n)$: mirror about $n=0$. Fold-then-delay $x(-(n-k)) = x(k-n)$: mirror, then place the old $n=0$ sample at $n=k$.
3. $x(2n)$ keeps every second sample (compressed, information lost); $x(n/2)$ inserts zeros. Mark $n=0$ with an arrow and label the heights.

Pole-zero plot with ROC

1. Draw the z-plane with the unit circle. Poles = $\boldsymbol{\times}$, zeros = $\boldsymbol{o}$ (write multiplicity). Remember extra poles/zeros at $z=0$ (or ∞) from $N-M$.
2. ROC is a circle, outside the circle, or a ring, bounded by poles. **Causal**: shade outside the outermost pole. **Anti-causal**: shade inside the innermost pole. **Two-sided**: the ring between two poles.
3. **No pole is ever inside the ROC.** Stable ⇔ the unit circle lies in the ROC. Finite sequences: all z except 0 and/or ∞.

Magnitude response from poles and zeros

1. Evaluate on the unit circle: $\boldsymbol{|H(e^{j\omega})| = |b_0| \cdot \Pi(\text{distance from } e^{j\omega} \text{ to each zero}) \,/\, \Pi(\text{distance to each pole})}$.
2. $\omega = 0$ is the point $z = 1$; $\omega = \pi$ is $z = -1$. Walking counter-clockwise: a pole near the circle at angle θ gives a **peak at $\boldsymbol{\omega \approx \theta}$**; a zero on the circle at θ gives a **null at $\boldsymbol{\omega = \theta}$**.
3. Notch: null at ω_0 with flat ≈ 1 elsewhere. Moving average: nulls at multiples of $2\pi/M$, lowpass-like with lobes. Oscillator: infinite peak (pole on the circle).

Aliasing picture (frequency axis)

1. Draw an axis in Hz with ticks at 0, $F_s/2$, F_s, $3F_s/2$ ... and fold it like a zig-zag at each multiple of $F_s/2$.
2. A tone at F lands at the same height on the first strip [0, $F_s/2$]: that is $F_{\text{alias}} = |F - \mathrm{k}F_s|$.
3. Sketch the samples: mark the sampling instants nT on the original sinusoid and connect them with the slower alias.

Impulse-response shape from the pole

1. Pole radius = decay: $|p| < 1$ decays, = 1 constant/oscillating, > 1 grows. Pole angle = oscillation: θ rad/sample ⇒ $F = \theta F_s/2\pi$.
2. Real positive pole: monotone. Real negative pole: alternates sign every sample. Complex pair: decaying cosine. Double pole: $n{\cdot}p^n$ (rises, peaks, decays).
3. First-order a^n: time constant $\tau = -1/\ln$ a; the envelope is 36.8 % of its start at $n = \tau$.

Block diagrams (DF-I, DF-II, FIR)

1. **FIR**: tapped delay line z^{-1} chain, a multiplier b_k per tap, one adder on the right.
2. **DF-II**: one delay chain $w(n-k)$. Left side adders form $w(n) = x(n) - \Sigma a_k w(n-k)$ (drawn gain = $-a_k$); right side forms $y(n) = \Sigma\, b_m w(n-m)$. Delays = $\max(M, N)$.
3. **DF-I**: input delays (b taps) feed an adder, then output delays (–a taps). Delays = $M + N$. Label every node, e.g. $x(n)$, $x(n-1)$, $y(n-1)$.

Sketching a convolution

1. Length $L_x + L_h - 1$, starts at (first index of x) + (first index of h). Total area $\Sigma y = \Sigma x \cdot \Sigma h$.
2. Two equal rectangles ⇒ triangle (peak = overlap length). Rectangle * exponential ⇒ rising then decaying. Causal exp * causal exp ⇒ starts at 0, tends to a smooth hump.
3. Per output point: fold h, slide it to n, multiply, sum. Draw the folded h under x and shade the overlap.

PART C · FORMULA SHEET

LECTURE 1 · SIGNALS & SAMPLING

Sampling relations

$$f = \frac{F}{F_s}$$

$x(n) = x_a(\mathrm{nT})$, $T = 1/F_s$. Distinct DT sinusoids: $-½ \leq f \leq ½$ ($-\pi \leq \omega \leq \pi$). Folding frequency $F_s/2$.

Quantization

$$\Delta = \frac{x_{\max} - x_{\min}}{L-1}, \quad L = 2^b, \quad P_q = \frac{\Delta^2}{12} \text{ (rounding)}$$

Truncation: mean-square error $\Delta^2/3$. Error $e_q(n) = x_q(n) - x(n)$, $|e_q| \leq \Delta/2$ for rounding.

SQNR

$$\mathrm{SQNR_{dB}} = 10\log_{10}\frac{P_x}{P_q}, \quad \text{full-scale sine: } 6.02b + 1.76$$

$P_x = (1/N)\Sigma x^2(n)$; a sinusoid has $A^2/2$.

Nyquist and aliasing

$$F_s \geq 2F_{\max} \qquad F_{\text{alias}} = F - \mathrm{k}F_s$$

Pick the integer k that puts F_{alias} in $[-F_s/2,\ F_s/2)$.

Quantize a sample

$$k = \mathrm{round}\left(\frac{x - x_{\min}}{\Delta}\right), \quad x_q = x_{\min} + k\Delta, \quad \Delta = \frac{x_{\max} - x_{\min}}{L-1}$$

Code = k in b-bit binary ($L = 2^b$). Error ≤ $\Delta/2$. Same as $L = (x_{\max} - x_{\min})/\Delta + 1$ solved for Δ; the +1 is outside the fraction.

Energy and average power

$$E = \Sigma|x(n)|^2, \quad P = \lim \frac{1}{2N+1}\Sigma_{-N}^{N}|x(n)|^2$$

Periodic: $P = (1/N)\Sigma$ over one period. Energy signal: $0 < E < \infty$ ($P = 0$). Power signal: $0 < P < \infty$ (E = ∞).

DT sinusoid periodicity

$$x(n+N) = x(n) \Leftrightarrow f = \frac{k}{N} \text{ rational}$$

Sum/product of sinusoids: N = lcm of the periods.

Even / odd parts

$$x_e = ½[x(n) + x(-n)], \quad x_o = ½[x(n) - x(-n)]$$

LECTURE 2 · DT SYSTEMS

Elementary signals

$$\delta(n), \quad u(n) = \Sigma_{k\geq 0}\delta(n-k), \quad r(n) = n{\cdot}u(n), \quad a^n u(n), \quad e^{j\omega n}$$

$\delta(n) = u(n) - u(n-1)$. Sifting: $x(n) = \Sigma_k\, x(k)\,\delta(n-k)$.

Convolution sum

$$y(n) = \Sigma_{k=-\infty}^{\infty} x(k)h(n-k) = x(n) * h(n)$$

Causal: $h(n) = 0$ for $n < 0$. Commutative, associative, distributive. Cascade: $h_1 * h_2$. Parallel: $h_1 + h_2$.

Linearity and time invariance

$$T[a_1x_1 + a_2x_2] = a_1T[x_1] + a_2T[x_2] \quad T[x(n-k)] = y(n-k)$$

Operations

delay: $x(n-k)$ advance: $x(n+k)$ fold: $x(-n)$ down-sample: $x(\mathrm{Mn})$

Block diagram elements: adder, multiplier, constant multiplier, unit delay z^{-1}, unit advance z.

BIBO stability

$$\Sigma_n\,|h(n)| < \infty$$

FIR: always stable. IIR: depends on pole locations.

Cross- and autocorrelation

$$r_{xy}(l) = \Sigma_n\, x(n)y(n-l), \quad r_{xy}(l) = r_{yx}(-l), \quad r_{xx}(l) = r_{xx}(-l)$$

$r_{xx}(0)$ = energy. $r_{yy} = r_{hh} * r_{xx}$; $r_{yx} = h * r_{xx}$. $r_{xy}(l) = x(l) * y(-l)$.

LECTURE 3 · LCCDE

General form

$$y(n) = -\Sigma_{k=1}^{N} a_k y(n-k) + \Sigma_{m=0}^{M} b_m x(n-m)$$

Non-recursive (FIR) if all $a_k = 0$. Recursive has feedback. Order = N.

Characteristic equation

$$\lambda^N + a_1\lambda^{N-1} + \ldots + a_N = 0$$

Roots = system poles. Distinct: $y_h = \Sigma\, C_k \lambda_k{}^n$. Repeated (multiplicity m): $(C_1 + C_2 n + \ldots + C_m n^{m-1})\lambda^n$.

Total response

$$y = y_h + y_p = y_{zi} + y_{zs}$$

$y_{zs} = h * x$ (relaxed). y_{zi} is the response to initial conditions with $x = 0$. y_p ↔ steady state, y_h ↔ transient (when stable).

Particular-solution forms

$$K \;\mid\; \mathrm{K a}^n \;\mid\; K_0 n^M + \ldots + K_M \;\mid\; K_1 \cos\omega_0 n + K_2 \sin\omega_0 n$$

Input: constant, a^n, polynomial n^M, sinusoid. ⚠ If the input form is already a homogeneous mode, multiply by n.

Impulse response of a recursive system

$$h(n) = y_{zs}(n) \text{ for } x = \delta(n)$$

Use $y(0) = b_0$, $y(1) = b_1 - a_1 b_0$ … to fit constants.

Direct Form II

$$w(n) = x(n) - \Sigma a_k w(n-k), \quad y(n) = \Sigma b_m w(n-m)$$

Memory = $\max(M, N)$ delays (DF-I uses $M + N$). ⚠ Drawn feedback gains are $-a_k$.

Second-order section

$$y(n) = -a_1 y(n-1) - a_2 y(n-2) + b_0 x(n) + b_1 x(n-1) + b_2 x(n-2)$$

Building block for higher-order filters (cascade of SOS).

LECTURE 4 · Z-TRANSFORM

Definition

$$X(z) = \Sigma_{n=-\infty}^{\infty} x(n) z^{-n}$$

ROC = set of z with a finite sum. One-sided: $\Sigma_{n=0}^{\infty}$ (used with initial conditions).

Common pairs (ROC)

$$\delta(n) \leftrightarrow 1 \quad u(n) \leftrightarrow \frac{1}{1 - z^{-1}} \quad a^n u(n) \leftrightarrow \frac{1}{1 - a z^{-1}} \;\; |z|>|a|$$

$-a^n\, u(-n-1) \leftrightarrow 1/(1-az^{-1})$, $|z| < |a|$. $\mathrm{na}^n u(n) \leftrightarrow az^{-1}/(1-az^{-1})^2$. $\mathrm{na}^n u(n) \leftrightarrow az^{-1}/(1-az^{-1})^2$. $(n+1)a^n\, u(n) \leftrightarrow 1/(1-az^{-1})^2$.

Damped sinusoids

$$r^n \cos(\omega_0 n) u(n) \leftrightarrow \frac{1 - r\cos\omega_0\, z^{-1}}{1 - 2r\cos\omega_0\, z^{-1} + r^2 z^{-2}}$$

$$r^n \sin(\omega_0 n) u(n) \leftrightarrow \frac{r\sin\omega_0\, z^{-1}}{1 - 2r\cos\omega_0\, z^{-1} + r^2 z^{-2}}$$

$r = 1$ for undamped cos/sin. Poles at $re^{\pm j\,\omega_0}$.

Properties

$$x(n-k) \leftrightarrow z^{-k} X(z) \quad a^n x(n) \leftrightarrow X(a^{-1}z) \quad x(-n) \leftrightarrow X(z^{-1})$$

$$n \cdot x(n) \leftrightarrow -z\frac{\mathrm{dX}}{\mathrm{dz}} \quad x_1 * x_2 \leftrightarrow X_1 X_2$$

Also linearity, correlation $r_{x_0 x_2} \leftrightarrow X_1(z)\, X_2(z^{-1})$, windowing (product ↔ convolution of transforms). ⚠ Shift moves the whole expression (including u).

Initial value (causal)

$$x(0) = \lim_{z\to\infty} X(z)$$

System function

$$H(z) = \frac{Y(z)}{X(z)} = \frac{b_0 + b_1 z^{-1} + \ldots + b_M z^{-M}}{1 + a_1 z^{-1} + \ldots + a_N z^{-N}} = b_0 z^{N-M} \frac{\Pi(z - z_k)}{\Pi(z - p_k)}$$

Zeros z_k: $H = 0$. Poles p_k: $H = \infty$. FIR: all poles at 0. Purely recursive (all-pole): $b_1 \ldots b_M = 0$.

Stability and ROC

$$\text{BIBO stable} \Leftrightarrow \text{ROC} \ni |z| = 1 \quad \text{causal} \Leftrightarrow |z| > r_{\max} \quad \text{causal \& stable} \Leftrightarrow \text{all } |p_k| < 1$$

Partial-fraction residues

$$A_k = (1 - p_k z^{-1}) X(z)\big|_{z=p_k}$$

Conjugate poles ⇒ conjugate residues: $x(n) = 2|A||p|^n \cos(n\theta + \angle A)\, u(n)$. Repeated poles: differentiation property, $z/(z-p)^2 \leftrightarrow \mathrm{np}^{n-1} u(n)$.

One-sided shift

$$Z^+\{y(n-1)\} = z^{-1} Y(z) + y(-1)$$

$$Z^+\{y(n-2)\} = z^{-2} Y(z) + z^{-1} y(-1) + y(-2)$$

$Y(z)$ = ZIR(z) + $H(z)\, X(z)$. ZIR numerator comes only from initial conditions.

PART D · RECIPES

LECTURE 1 · SAMPLING & QUANTIZATION

Aliasing / what DT signal do I get?

1. Read each frequency F (Hz) from the argument: $\Omega = 2\pi F$.
2. $f = F/F_s$. If $|f| \le ½$ you are done: $x(n) = A\cos(2\pi\mathrm{fn} + \varphi)$.
3. Otherwise subtract the nearest integer: $f' = f - k$. The alias frequency is $F' = f' \cdot F_s$.
4. Reconstruction (ideal D/A) outputs only $|F| \le F_s/2$, so you get the alias, not the original.

cos is even (alias keeps its sign); sin is odd (alias flips sign when f' is negative). Add tones at equal frequency after aliasing.

Quantizer sizing and SQNR

1. $L = (x_{\max} - x_{\min})/\Delta + 1$ and $b = \lceil \log_2 L \rceil$ (or Δ = range/$(L-1)$ from b).
2. Rounding: $P_q = \Delta^2/12$. Truncation (mean-square): $\Delta^2/3$.
3. $P_x = A^2/2$ for a sinusoid, or $(1/N)\Sigma x^2$.
4. SQNR = $10\log(P_x/P_q)$; full-scale sine ⇒ $6.02\,b + 1.76$ dB.

Every extra bit = +6.02 dB. State which Δ convention (range/$(L-1)$ vs range/L) you used.

Quantize one sample (level, code, error)

1. $\Delta = (x_{\max} - x_{\min})/(L-1)$, with $L = 2^b$.
2. k = round$((x - x_{\min})/\Delta)$. Clip to $0 \ldots L-1$ if x is outside the range.
3. $x_q = x_{\min} + k\Delta$. Code = k in b-bit binary.
4. Error = $x_q - x$, and $|\text{error}| \le \Delta/2$ for rounding.

Example: range $-1 \ldots 1$, $L = 8$, $x = 0.30$: $\Delta = 2/7$, k = round$(1.3 \cdot 3.5)$ = round$(4.55) = 5$, $x_q = 0.429$, code 101, error 0.129 ($\le \Delta/2 = 0.143$). Truncation would use the floor instead of round.

Periodicity of a DT sinusoid

1. $f = \omega_0/2\pi$.
2. Rational (k/N in lowest terms)? Yes → fundamental period N. Irrational → not periodic.
3. Sum or product of sinusoids: N = lcm of the individual periods.

LECTURE 2 · DT SIGNALS & SYSTEMS

Shifting, folding and ordering

1. Write the final formula with n substituted into $x(\cdot)$: e.g. fold-then-delay-by-4 → $x(-(n-4)) = x(4-n)$.
2. Build a table of n against the argument of x, then read off $x(\text{argument})$.
3. Delay by k = shift right ($x(n-k)$); advance = shift left ($x(n+k)$).

Fold-then-delay ≠ delay-then-fold. Down-sampling $x(\mathrm{Mn})$ is time-variant.

Classify a system (static, linear, TI, causal, stable)

1. **Static?** Output depends only on $x(n)$ at the same n.
2. **Causal?** Every index in $x(\cdot)$ is ≤ n. Look at the worst case, e.g. $x(n^2+1)$, $x(4n)$, $x(-n)$ can reach the future.
3. **Linear?** Test $T[a_1 x_1 + a_2 x_2] = a_1 T[x_1] + a_2 T[x_2]$. Quick fail: zero in must give zero out.
4. **Time-invariant?** Feed $x(n-k)$: compare $T[x(n-k)]$ with $y(n-k)$. Different ⇒ time-variant (an explicit n multiplying the input, or n scaled inside $x(\cdot)$).
5. **BIBO stable?** Try a bounded input (e.g. $x = 1$): does y stay bounded?

Convolution table

1. Length of $y = L_x + L_h - 1$; first index = sum of first indices.
2. For each n: fold h, shift by n, multiply with x sample by sample, sum.
3. Check: $\Sigma y = \Sigma x \cdot \Sigma h$.
4. Alternatively, treat as polynomial multiplication of the coefficient lists.

Correlation

1. $r_{xy}(l) = \Sigma\, x(n)\, y(n-l)$: shift, multiply, sum (no fold).
2. $r_{xy}(l) = r_{yx}(-l)$; r_{xx} is even and peaks at $l = 0$ (= energy).
3. Through an LTI system: $r_{yy} = r_{hh} * r_{xx}$ and $r_{yx} = h * r_{xx}$.
4. To find a delay D buried in noise, find the lag where r_{yx} peaks.

LECTURE 3 · LCCDE

Direct method for y(n) = $y_h + y_p$

1. **Characteristic equation:** replace $y(n-k)$ by λ^{-k} and solve for λ.
2. y_h**:** distinct roots → $\Sigma C_k \lambda_k{}^n$; repeated root → $(C_1 + C_2 n)\lambda^n$; complex pair → $r^n(A\cos n\theta + B\sin n\theta)$.
3. y_p**:** same form as the input (K for a step, Ka^n for a^n, $K_1\cos + K_2\sin$ for sinusoids). If the input matches a root, multiply by n.
4. Compute $y(0)$, $y(1)$, … **by recursion** using the initial conditions.
5. Fit the C's to $y(0)$, $y(1)$ of the **total** solution $y_h + y_p$.

ZIR = response to initial conditions with $x = 0$. ZSR = response to the input with zero initial conditions (convolution with h). Total = ZIR + ZSR.

Recursion by hand

1. Rearrange: $y(n) = -\Sigma a_k y(n-k) + \Sigma b_m x(n-m)$.
2. Start at $n = 0$ with the given $y(-1)$, $y(-2)$, … (zero if relaxed).
3. For $h(n)$ set $x = \delta(n)$; the input terms vanish after $n = M$ and $h(n)$ follows the feedback alone.

Draw Direct Form I / II

1. **Normalize** so the coefficient of $y(n)$ is 1 (divide the whole equation).
2. DF-I: one delay line for x (M delays) and one for y (N delays); $y(n) = \Sigma\ b_m x(n-m) - \Sigma\ a_k y(n-k)$.
3. DF-II: $w(n) = x(n) - \Sigma a_k w(n-k)$, $y(n) = \Sigma b_m w(n-m)$; one shared line of $\max(M, N)$ delays.
4. Feedback multipliers in the drawing are $-a_k$; feedforward are b_m.
5. FIR = only the b taps (no feedback).

Check stability on the poles before realizing: roots of $\lambda^N + a_1\lambda^{N-1} + \ldots + a_N = 0$ must be inside the unit circle.

Read a block diagram

1. Label the output of every delay ($w(n-1)$, $w(n-2)$, …).
2. Write each adder as an equation exactly as drawn.
3. Eliminate the internal signal w to get the y–x relation.
4. Convert to the standard a_k/b_m form last.

LECTURE 4 · Z-TRANSFORM

Z-transform and ROC

1. Finite: write $X(z) = \Sigma x(n) z^{-n}$ term by term; ROC = whole plane minus 0 and/or ∞ depending on which powers appear.
2. $a^n u(n) \to 1/(1-az^{-1})$, $|z| > |a|$. $-a^n u(-n-1) \to$ same expression, $|z| < |a|$.
3. Sum of terms: ROC is the **intersection**; if empty, no Z-transform.
4. Use the property table (shift, scale, differentiate, convolve) instead of summing again.

ROC never contains a pole. Right-sided = outside, left-sided = inside, two-sided = ring.

Inverse Z (the decision tree)

1. **Improper?** (numerator degree ≥ denominator degree in z^{-1}) divide first: polynomial part → impulses.
2. **Factor** the denominator into $(1 - p_k z^{-1})$.
3. **Residues:** distinct p_k: $A_k = [(1 - p_k z^{-1})X(z)]$ at $z = p_k$. Repeated: expand $X(z)/z$ in $z/(z-p)^m$ terms. Complex pair: A and A*.
4. **Pick each term's side from the ROC:** pole inside the ROC's inner edge → right-sided $p^n u(n)$; pole outside → left-sided $-p^n u(-n-1)$.
5. Check $x(0)$ and $x(1)$ with long division.

From difference equation to H(z)

1. Replace $y(n-k) \to z^{-k} Y(z)$, $x(n-m) \to z^{-m} X(z)$.
2. $H(z) = Y/X = (b_0 + b_1 z^{-1} + \ldots)/(1 + a_1 z^{-1} + \ldots)$.
3. Multiply top and bottom by z^N to read poles/zeros as roots in z.
4. $h(n) = Z^{-1}\{H\}$; step response: invert $H(z)/(1-z^{-1})$.
5. Stable causal ⇔ all poles $|p| < 1$.

Nonzero initial conditions (one-sided Z)

1. Use $Z^+\{y(n-1)\} = z^{-1}\,Y + y(-1)$; $Z^+\{y(n-2)\} = z^{-2}\,Y + z^{-1}y(-1) + y(-2)$.
2. Collect: $Y(z)\cdot D(z) = N_{ic}(z) + N_{input}(z)X(z)$.
3. ZIR = N_{ic}/D; ZSR = $H(z)X(z)$.
4. Invert each part with the inverse-Z tree.

DESIGN PROBLEMS

Pole-zero placement

1. Convert the frequency to $\omega_0 = 2\pi F/F_s$ (rad/sample).
2. **To remove ω_0:** zeros at $e^{\pm j\omega_0}$ (numerator $1 - 2\cos\omega_0\ z^{-1} + z^{-2}$). Add poles at $r\cdot e^{\pm j\omega_0}$ with r close to 1 to keep neighbors untouched; width ≈ $2(1-r)$ rad.
3. **To pass ω_0 (resonate):** poles near $e^{\pm j\omega_0}$; $r = 1$ gives an oscillator (marginal).
4. **Smoothing:** real pole at $a \approx 1$; time constant ≈ −1/ln a samples.
5. Write $H(z)$ → difference equation → coefficients.
6. Verify: poles inside the circle, DC gain $H(1)$, and the frequency response at ω_0 and a neighbor.

Sizing a converter

1. $F_s = 2B \times (1 + \text{guard})$ and filter: pass B, stop $F_s - B$.
2. Bits: from the Δ spec (L = range/Δ + 1) or the SQNR spec ($b \geq (\text{SQNR} - 1.76)/6.02$).
3. Bit rate = $F_s \cdot b$.
4. Report the trade-offs (filter difficulty vs rate).

PART E · RED FLAGS

Red flags (the traps that cost points)

- **Aliasing:** subtract the integer *before* scaling by F_s; sin flips sign, cos does not. A DT sinusoid is periodic only if f is *rational*.
- **Fold/delay order:** fold-then-delay by k is $x(-(n-k))$; delay-then-fold is $x(-n-k)$. Different signals.
- **Linearity:** $y = 2x + 3$ is *not* linear (zero in ≠ zero out) but is time-invariant. $y = x^2$ is non-linear. $y = x(4n)$ is time-variant. $y = n{\cdot}x(n-1)$ is time-variant *and* unstable.
- **Convolution:** length = $L_x + L_h - 1$; first index = sum of first indices; $\Sigma y = \Sigma x \Sigma h$. Don't forget zero taps are still part of $h(n)$.
- **Recursive ≠ IIR.** A recursive running sum can be FIR (pole–zero cancellation); FIR ⇔ finite $h(n)$.
- **LCCDE:** fit C_1, C_2 with $y(0)$, $y(1)$ of the *total* solution; resonance (input = natural mode) ⇒ multiply y_p by n; repeated root ⇒ extra factor n.
- **Realization:** normalize the $y(n)$ coefficient to 1 first; drawn feedback gain is $-a_k$; DF-II uses $\max(M, N)$ delays, DF-I uses $M + N$.
- **Z-transform shift:** $a^n u(n-2) \neq a^{n-2} u(n-2)$ — rewrite before using z^{-2}. Time reversal flips the ROC to the inside.
- **ROC:** never contains a pole. $X(z)$ without ROC does not define $x(n)$. Sum of right- and left-sided pieces may have *empty* ROC (no transform).
- **Improper $X(z)$:** divide first; the polynomial part gives impulses at the start of $x(n)$.
- **Stability:** BIBO stable ⇔ unit circle in the ROC. Causal ⇒ all poles inside $|z| < 1$. A pole at 2 can give a stable system only if non-causal.
- **Pole angle ↔ frequency:** $\omega_0 = 2\pi F/F_s$. Radius ↔ decay; $r = 1$ ⇒ sustained oscillation (marginal); $r > 1$ ⇒ growth.

Numbers worth knowing cold

- SQNR(full-scale sine) = $6.02b + 1.76$ dB; every bit adds 6.02 dB; truncation is 6.02 dB worse than rounding (mean-square).
- 2cos 36° = 1.618 (golden ratio), sin 36° = 0.5878, cos 45° = sin 45° = 0.7071, cos 60° = 0.5, sin 60° = 0.866.
- $\Sigma_{k=0}{}^n\ a^k = (1 - a^{n+1})/(1 - a)$; $\Sigma_{k=0}{}^{\infty}\ a^k = 1/(1 - a)$ for $|a| < 1$.
- Step response of $1/(1 - az^{-1})$: $(1 - a^{n+1})/(1 - a)$; settles (95 %) in about $3/(-\ln a)$ samples.
- Moving average of length M: zeros at $\omega = 2\pi k/M$, DC gain 1, $h = 1/M$ each.

PART F · DESIGN PROBLEMS (WORKED)

DS-01 · ADC front end for speech

Spec: *speech limited to $B = 4$ kHz. The A/D must give SQNR ≥ 60 dB for a full-scale sinusoid, and the anti-aliasing filter must reject by ≥ 60 dB anything that would fold into 0–4 kHz. Keep the bit rate modest. Choose F_s, the anti-alias filter edges and the number of bits, and report the bit rate.*

1. **Minimum rate:** Nyquist says $F_s \geq 2B = 8$ kHz. But then the filter must pass 4 kHz and block everything above $F_s - B = 4$ kHz: a brick wall (zero transition band) — not realizable.
2. **Pick a guard band:** $F_s = 10$ kHz. A tone at F between $F_s/2$ and F_s folds to $F_s - F$, so everything above $F_s - B = 6$ kHz lands in the 0–4 kHz band (6 kHz → 4 kHz, 7 kHz → 3 kHz, …). So the filter needs **pass edge 4 kHz, stop edge 6 kHz, ≥ 60 dB**: a 2 kHz-wide transition band.
3. **Bits:** SQNR ≈ $6.02b + 1.76$ dB ≥ 60 ⇒ $b \geq 9.7$ ⇒ $\boldsymbol{b = 10}$ **bits** (SQNR = 61.96 dB).
4. **Bit rate:** $R = F_s{\cdot}b = 10\,000 \times 10$ = **100 kb/s**.

F_s	Transition band	Bit rate (10 b)	Verdict
8 kHz	4 → 4 kHz (none)	80 kb/s	impossible filter
10 kHz	4 → 6 kHz	**100 kb/s**	practical compromise
16 kHz	4 → 12 kHz	160 kb/s	easy filter, wasteful rate

Real telephony makes the same compromise: voice is band-limited to ≈ 3.4 kHz and sampled at 8 kHz (transition 3.4 → 4.6 kHz), then μ-law companded to 8 bits for 64 kb/s. The 6 dB/bit rule holds only for a *full-scale* signal; quiet speech gets a poorer SQNR, which is exactly why companding exists.

DS-02 · Sensor digitizer from a resolution spec

Spec: *a sensor output spans ±2 V with 2.4 kHz bandwidth. The step size must be no coarser than 1 mV. (a) How many levels and bits? (b) Choose F_s with 25 % margin over Nyquist. (c) What is the bit rate? (d) What SQNR does a full-scale sine achieve?*

1. **(a)** Using the slide's convention $\Delta = (x_{max} - x_{min})/(L-1)$: $L - 1 \geq 4\ V / 1$ mV = 4000 ⇒ $L \geq 4001$ ⇒ $2^{11} = 2048$ is too few, $2^{12} = 4096$ works ⇒ **12 bits**. Actual $\Delta = 4/4095$ = **0.977 mV** ≤ 1 mV ✓. (11 bits would give $\Delta = 4/2047 = 1.95$ mV ×.)
2. **(b)** Nyquist = 2 × 2.4 kHz = 4.8 kHz; × 1.25 ⇒ $\boldsymbol{F_s = 6}$ **kHz**. Anti-alias filter: pass 2.4 kHz, stop by $F_s - 2.4 = 3.6$ kHz.
3. **(c)** $R = 6000 \times 12$ = **72 kb/s**.
4. **(d)** Rounding: $P_q = \Delta^2/12 = 7.95{\times}10^{-8}\ V^2$; sine $P_x = 2^2/2 = 2\ V^2$ ⇒ SQNR = $10\log(2/7.95{\times}10^{-8})$ = **74.0 dB** (= $6.02{\cdot}12 + 1.76$ ✓).

Notice the order of design: resolution spec → bits → SQNR falls out. The reverse (SQNR spec → bits) was DS-01.

DS-03 · Design an inverse system (echo / ISI equalizer)

Spec: *a channel adds one echo, $y(n) = x(n) + 0.5x(n-1)$. Design a causal, stable system G that recovers $x(n)$ from $y(n)$. Give $g(n)$ and its difference equation, and prove $G * H = \delta$. Then repeat for a stronger echo $y(n) = x(n) + 2x(n-1)$.*

1. Channel: $h(n) = \delta(n) + 0.5\,\delta(n-1) \Rightarrow H(z) = 1 + 0.5\,z^{-1}$. The equalizer must satisfy $G(z)\,H(z) = 1 \Rightarrow$ $\boldsymbol{G(z) = 1/(1 + 0.5z^{-1})}$.
2. Pole at $z = -0.5$, inside the unit circle ⇒ causal and stable: $\boldsymbol{g(n) = (-0.5)^n u(n)}$ (IIR). Difference equation: $\boldsymbol{x(n) = y(n) - 0.5\,x(n-1)}$ (recursive — it subtracts its own previous estimate).
3. **Proof:** $g * h = (-0.5)^n u(n) + 0.5(-0.5)^{n-1}u(n-1) = (-0.5)^n[u(n) - u(n-1)] = \boldsymbol{\delta(n)}$ ✓.
4. **Strong echo:** $H(z) = 1 + 2z^{-1}$ has a zero at $z = -2$ (outside the unit circle). The inverse $1/(1 + 2z^{-1})$ has a pole at −2: the causal inverse $(-2)^n u(n)$ **explodes**; the stable inverse is the anti-causal $-(-2)^n u(-n-1)$, which needs future samples of y (usable offline, or with delay).

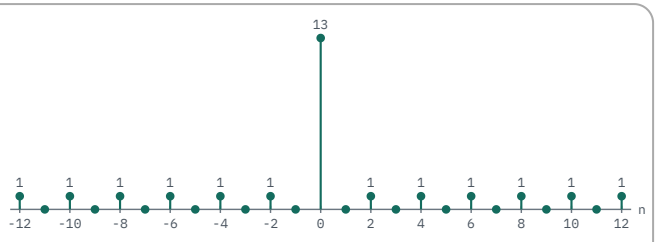
⚠ A causal, stable inverse exists only if all zeros of $H(z)$ lie inside the unit circle (a minimum-phase channel). This is the DSP version of the ISI/equalizer idea from digital communications.

DS-04 · Radar delay estimation with a Barker code

Spec: *a radar transmits the Barker-13 sequence $x(n) = \{1, 1, 1, 1, 1, -1, -1, 1, 1, -1, 1, -1, 1\}$. The receiver sees $y(n) = a{\cdot}x(n - D) + v(n)$, v white noise. (a) With $F_s = 1$ MHz and a correlation peak at lag $D = 20$, what is the range? (b) What F_s is needed to separate targets 15 m apart? (c) What are the peak and sidelobe levels of the code, and the noise benefit of correlating? ($c = 3{\times}10^8$ m/s)*

1. **Detector:** compute $r_{yx}(l) = \Sigma\ y(n)\,x(n-l)$. The peak occurs at $l = D$ (with the lecture's $r_{xy}(l) = \Sigma\ x(n)y(n-l)$ it appears at $l = -D$ — same information, folded).
2. **(a)** Delay = $D{\cdot}T = 20/1$ MHz = 20 µs. The wave goes out *and* back, so $\boldsymbol{R = c{\cdot}t_d/2 = 3{\times}10^8 \times 20{\times}10^{-6}\ /\ 2}$ $\boldsymbol{= 3000\ m}$.

3. **(b)** Resolution is one lag: $\Delta R = c\cdot T/2 \le 15\ m \Rightarrow T \le 0.1\ \mu s \Rightarrow$ **$F_s \ge 10$ MHz**.
4. **(c)** Autocorrelation of Barker-13 (figure): peak = 13 at $l = 0$, sidelobes only 0 or ±1. Peak-to-sidelobe = 13:1 ⇒ 20log 13 = **22.3 dB**.
5. **Noise benefit:** at the right lag the 13 aligned samples add coherently, giving a peak of 13a (power $169a^2$), while zero-mean white noise adds incoherently (power $13\sigma^2$). SNR improves from a^2/σ^2 to $13a^2/\sigma^2$: a gain of 13, *i.*e. **10log 13 = 11.1 dB** (the matched-filter gain).



$r_xx(l)$ of the Barker-13 sequence: one sharp peak of 13 at $l = 0$; every other lag is 0 or 1.

Why Barker? For any other length-13 ±1 code the sidelobes would usually be larger. Low sidelobes prevent a strong nearby echo from masking a weak distant one.

DS-05 · Exponential smoother from a settling-time spec

Spec: *design $y(n) = a\cdot y(n-1) + (1-a)\cdot x(n)$ so that its unit-step response reaches 95 % of its final value by $n = 20$. Find a, the DC gain, the time constant, and the noise-variance reduction for white input noise.*

1. **Step response by the direct method:** $y_h = Ca^n$ (the pole is a); for $x = u(n)$, $y_p = K$ with $K(1-a) = 1-a \Rightarrow K = 1$. Relaxed: $y(0) = 1 - a \Rightarrow C = -a$. So **$y(n) = 1 - a^{n+1}$** for $n \ge 0$.
2. **Spec:** $1 - a^{21} \ge 0.95 \Rightarrow a^{21} \le 0.05 \Rightarrow a \le 0.05^{1/21} = e^{\ln 0.05/21} = e^{-0.1427} \Rightarrow$ **$a \approx 0.867$**.
3. **DC gain:** $H(1) = (1-a)/(1-a) =$ **1** (final value of the step is 1, so no scaling needed).
4. **Time constant:** $\tau \approx -1/\ln a = 7.0$ samples; 95 % ≈ $3\tau = 21$ samples ✓. Pole at $z = a = 0.867$ (inside the unit circle: stable).
5. **Noise:** $h(n) = (1-a)\,a^n \Rightarrow \Sigma h^2 = (1-a)^2/(1-a^2) = (1-a)/(1+a) = 0.0712 \Rightarrow$ white-noise variance is reduced to **7 %** (−11.5 dB) at the price of a 21-sample lag.

The design knob is a single pole location. $a \to 1$ gives more smoothing but slower response; $a \to 0$ gives no smoothing ($y = x$). That trade-off between noise and speed is the whole story of low-pass design.

DS-06 · Echo/reverb effect and its memory cost

Spec: *audio at $F_s = 8$ kHz. Add an echo 0.25 s after the sound at half amplitude, and let each echo re-echo. Give the difference equation, $h(n)$, $H(z)$, the stability condition and the memory required.*

1. **Delay in samples:** $D = 0.25\ s \times 8000 =$ **2000**.
2. **Equation:** the output feeds back through a 2000-sample delay with gain 0.5: **$y(n) = x(n) + 0.5\,y(n-2000)$**.
3. **$h(n)$:** $h(n) = 0.5\cdot h(n-2000) + \delta(n) \Rightarrow$ **$h(n) = \Sigma_{k\ge 0}\,(0.5)^k\,\delta(n-2000k)$**: impulses of height 1, ½, ¼, … every 2000 samples ⇒ IIR (infinite length from only two coefficients).
4. **$H(z)$** = $1/(1-0.5z^{-2000})$: 2000 poles on a circle of radius $0.5^{1/2000} =$ **0.99965** — just inside the unit circle ⇒ stable. In general stability needs $|g| < 1$ for feedback gain g.
5. **Memory:** a Direct Form II realization shares one delay line: **2000 delay elements** (= 0.25 *s* of audio). A feedforward FIR with the same echoes truncated at K repeats would need 2000K.
6. **DC gain:** $1/(1-0.5) = 2 \Rightarrow$ steady tones come out twice as loud; scale the input by 0.5 if unity DC gain is required.

⚠ Gain $g = 1$ would repeat the echo forever (poles on the unit circle, marginal). $g > 1$ grows without bound. Always check $|g| < 1$.

PART G · SOLVED PROBLEMS (WORKED, WITH FIGURES)

L1-01 · Sampling and aliasing of one cosine

Given $x_a(t) = 3\cos(100\pi t)$. (a) Find the minimum sampling rate that avoids aliasing. (b) The signal is sampled at $F_s = 200$ Hz — what DT signal results? (c) It is sampled at $F_s = 75$ Hz instead — what DT signal results? (d) Find the frequency $0 < F < F_s/2$ of a sinusoid that yields samples identical to those in (c).

1. Read the frequency: $\Omega = 100\pi = 2\pi F \Rightarrow$ **$F = 50$ Hz**.
2. (a) Nyquist: $F_s \ge 2F \Rightarrow$ **$F_s \ge 100$ Hz**.
3. (b) $f = F/F_s = 50/200 = 1/4$, inside $|f| \le ½$, so no aliasing: **$x(n) = 3\cos(\pi n/2)$**.
4. (c) $f = 50/75 = 2/3 > ½ -$ aliased. $x(n) = 3\cos(4\pi n/3) = 3\cos(2\pi n - 2\pi n/3) =$ **$3\cos(2\pi n/3)$**.
5. (d) Subtract the integer: $f_{alias} = 2/3 - 1 = -1/3$, so $|f| = 1/3 \Rightarrow F = (1/3)(75) =$ **25 Hz**. Check: $3\cos(2\pi\cdot 25\cdot n/75) = 3\cos(2\pi n/3)$ ✓.

⚠ Folding frequency is $F_s/2 = 37.5$ Hz; the 50 Hz tone is above it and "folds" down to 25 Hz. For a *sine* the alias flips sign (sin is odd); for cosine it does not.

L1-02 · Sampling range and folding of a band-limited signal

An analog signal contains frequencies up to 10 kHz. (a) What range of sampling frequencies allows exact reconstruction? (b) The signal is sampled at $F_s = 8$ kHz. Examine what happens to the components $F_1 = 5$ kHz and $F_2 = 9$ kHz.

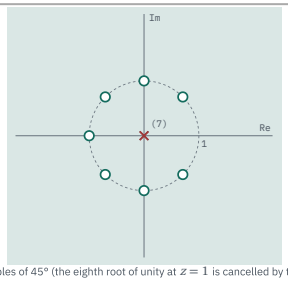
1. (a) **$F_s \ge 2\cdot 10$ kHz = 20 kHz** (no mathematical upper limit; higher costs more samples).
2. (b) Folding frequency = $F_s/2 = 4$ kHz, so only −4…+4 kHz is distinguishable.
3. $F_1 = 5$ kHz: $f = 5/8$; subtract $1 \to -3/8 \Rightarrow$ appears at **3 kHz**.
4. $F_2 = 9$ kHz: $f = 9/8$; subtract $1 \to 1/8 \Rightarrow$ appears at **1 kHz**.

$$F_{alias} = F - k\cdot F_s,\ \text{choose integer k so that } |F_{alias}| \le F_s/2$$

DS-07 · Moving-average filter that nulls 50 Hz hum

Spec: *a signal sampled at 400 Hz is polluted by 50 Hz mains hum. Design a causal FIR moving-average filter with unity DC gain that completely removes it. Give M, $h(n)$, $H(z)$ and the zero locations, and state the other frequencies it also removes.*

1. **Hum frequency:** $\omega = 2\pi\cdot 50/400 =$ **$\pi/4$**.
2. A length-M moving average has $H(z) = (1/M)\Sigma z^{-k} = (1/M)(1-z^{-M})/(1-z^{-1})$. Its zeros are the M-th roots of unity except $z = 1$ (cancelled by the denominator), *i.*e. at $\omega = 2\pi k/M$, $k = 1 \ldots M-1$.
3. Make the first zero land on $\pi/4$: $2\pi/M = \pi/4 \Rightarrow$ **$M = 8$**.
4. **$h(n) = 1/8$ for $0 \le n \le 7$** (unity DC gain: $\Sigma h = 1$). **$H(z) = (1/8)(1-z^{-8})/(1-z^{-1})$**.
5. **Also nulled:** $k = 2, 3, 4 \Rightarrow$ 100, 150, 200 Hz (every multiple of $F_s/M = 50$ Hz). Gain at 25 Hz is 0.64 and at DC is 1.
6. **Implementation:** direct FIR: $y(n) = (1/8)\Sigma_{k=0}^{7} x(n-k)$ (7 adds, 7 delays), or recursive running sum $y(n) = y(n-1) + [x(n) - x(n-8)]/8$ (same output; the pole at $z = 1$ is cancelled by a zero, so errors can accumulate in finite precision).



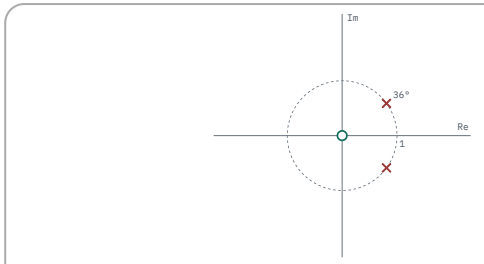
Seven zeros on the unit circle at multiples of 45° (the eighth root of unity at $z = 1$ is cancelled by the pole there), and seven poles at the origin. The zero at ±45° removes the 50 Hz hum.

The cost: a 25 Hz component loses 36 % of its amplitude, and the filter is a crude low-pass. DS-09 shows how to remove one frequency without damaging its neighbors.

DS-08 · Digital sine oscillator at 2 kHz

Spec: *design a second-order oscillator producing a sinusoid of peak amplitude 1 and frequency 2000 Hz at $F_s = 20$ kHz. (a) Find $H(z)$ and the difference equation, with the oscillation started by an impulse. (b) If instead no impulse is available, what initial conditions $y(-1)$, $y(-2)$ start the same oscillation? (c) If $y(-1) = 0$, what $y(-2)$ gives a unit-amplitude sinusoid? (d) Repeat (a) and (b) for a cosine start.*

1. **Frequency:** $\omega_0 = 2\pi\cdot 2000/20000 =$ **$\pi/5 = 36°$**. We want $y(n) = \sin(\omega_0 n)\,u(n)$ as the impulse response: $Z\{\sin(\omega_0 n)u(n)\} = \sin\omega_0\, z^{-1}/(1 - 2\cos\omega_0\, z^{-1} + z^{-2})$.
2. **(a) $H(z) =$ 0.5878 $z^{-1}/(1 -$ 1.618 $z^{-1} + z^{-2})$**, poles at $e^{\pm j36°}$ (on the unit circle). Difference equation: **$y(n) = 1.618\,y(n-1) - y(n-2) + 0.5878\,x(n-1)$**. Drive with $x = \delta(n)$. (2cos 36° = 1.618 is the golden ratio.)
3. **(b) Free-running ($x = 0$):** $y(n) = 1.618\,y(n-1) - y(n-2)$. To reproduce $\sin(\omega_0 n)$, $n \ge 0$, the "history" must be the sine extended backwards: **$y(-1) = \sin(-\omega_0) = -0.5878$, $y(-2) = \sin(-2\omega_0) = -0.9511$**. Check: $y(0) = 1.618(-0.5878) + 0.9511 = 0$ ✓; $y(1) = 0 - (-0.5878) = 0.5878 = \sin 36°$ ✓.
4. **(c) With $y(-1) = 0$:** $y(0) = -y(-2)$, $y(1) = -1.618\,y(-2)$, … = $-y(-2)\cdot\sin((n+1)\omega_0)/\sin\omega_0$. For unit amplitude choose **$y(-2) = -\sin\omega_0 = -0.5878$**: then $y(n) = \sin((n+1)\omega_0)$ — the same 2 kHz sine, advanced by one sample.
5. **(d) Cosine version** (peak 1, starts at 1): want $y(n) = \cos(\omega_0 n)\,u(n) \leftrightarrow (1 - \cos\omega_0\, z^{-1})/(1 - 2\cos\omega_0\, z^{-1} + z^{-2})$. So **$H(z) = (1 - 0.809z^{-1})/(1 - 1.618z^{-1} + z^{-2})$**, $y(n) = 1.618\,y(n-1) - y(n-2) + x(n) - 0.809\,x(n-1)$, impulse input. Free-running, the history is the cosine extended backwards: **$y(-1) = \cos\omega_0 = 0.809$, $y(-2) = \cos 2\omega_0 = 0.309$**.
6. **Why $y(-1) = 0$ cannot give a cosine:** with $y(-1) = 0$ and $x = 0$ the one-sided transform is $Y^+(z) = -y(-2)/(1 - 1.618z^{-1} + z^{-2})$, which is a *sine* shape: $y(n) = -y(-2)\cdot\sin((n+1)\omega_0)/\sin\omega_0$. For general preloads the numerator is $1.618\,y(-1) - y(-2) - y(-1)\,z^{-1}$; matching the cosine's s $1 - 0.809\,z^{-1}$ forces $y(-1) = 0.809$ and then $y(-2) = 1.618\cdot 0.809 - 1 = 0.309$. With $y(-2) = -1$ you get $\sin((n+1)\omega_0)/\sin\omega_0$, whose peak is 1/sin 36° = 1.70, not 1.



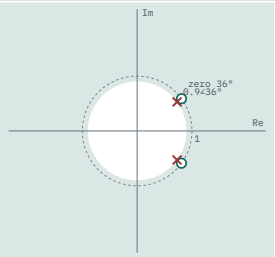
Poles at ±36° on the unit circle: marginally stable, so the sine neither decays nor grows. The numerator z^{-1} puts one zero at the origin.

⚠ Poles exactly on the unit circle are not BIBO stable. In finite-precision arithmetic the coefficient 1.618 is rounded, moving the poles slightly inside (decaying tone) or outside (growing tone). Real oscillators use extra amplitude control.

DS-09 · Second-order notch filter for 60 Hz

Spec: *remove 60 Hz hum from a signal sampled at $F_s = 600$ Hz while leaving other frequencies almost untouched. Place the poles and zeros, give $H(z)$, the difference equation, the DC gain, and the notch width.*

1. **Notch frequency:** $\omega_0 = 2\pi\cdot 60/600 =$ **$\pi/5 = 36°$**, 2cos $\omega_0 = 1.618$.
2. **Zeros** exactly on the unit circle at $e^{\pm j\omega_0}$ give an exact null: numerator **$1 - 1.618\,z^{-1} + z^{-2}$**.
3. **Poles** at $r\cdot e^{\pm j\omega_0}$, just inside the circle, cancel the zeros' effect everywhere except right at ω_0. Take $r = 0.9$: denominator **$1 - 2r\cos\omega_0\, z^{-1} + r^2 z^{-2} = 1 - 1.4562\,z^{-1} + 0.81\,z^{-2}$**.
4. **$H(z) = (1 - 1.618z^{-1} + z^{-2})/(1 - 1.4562z^{-1} + 0.81z^{-2})$**. Difference equation: **$y(n) = 1.4562\,y(n-1) - 0.81\,y(n-2) + x(n) - 1.618\,x(n-1) + x(n-2)$**.
5. **DC gain:** $H(1) = (2 - 1.618)/(1 - 1.4562 + 0.81) = 0.382/0.3538 =$ **1.080**. To make it unity, scale the numerator by $1/1.080 = 0.926$.
6. **Width:** the −3 dB points are about $\pm(1 - r) = \pm 0.1$ rad/sample either side of ω_0, so the notch width ≈ $2(1 - r) = 0.2$ rad ≈ 19 Hz. Numerically $|H|$ at $\omega_0 \pm 0.1$ is $0.762 \approx 0.707 \times 1.080$ ✓.
7. **Stability:** poles at radius $0.9 < 1$ ✓. Trade-off: $r \to 1$ gives a narrower notch but a longer transient (about $1/(1-r) = 10$ samples to settle).



Zeros on the unit circle at ±36° (exact null at 60 Hz); poles at the same angle but radius 0.9 (pulls the response back up on either side). ROC $|z| > 0.9$ includes the unit circle.

General recipe: zero at $e^{j\omega_0}$ kills ω_0; a pole just beside it restores the neighbors. Equivalent for a resonator (peak): put the *pole* near the circle and the zero far away.

Both aliases land inside the 0–4 kHz band, indistinguishable from genuine 3 kHz and 1 kHz tones. This is why an anti-aliasing filter must come *before* the sampler.

L1-03 · Three-tone signal: Nyquist rate, sampled sequence, reconstruction

$x_a(t) = 3\cos(2000\pi t) + 5\sin(6000\pi t) + 10\cos(12000\pi t)$. (a) What is the Nyquist rate? (b) What DT signal results from sampling at $F_s = 5000$ samples/s? (c) What analog signal $y_a(t)$ is reconstructed from those samples by an ideal D/A converter?

1. Frequencies: $2000\pi \to$ **1 kHz**; $6000\pi \to$ **3 kHz**; $12000\pi \to$ **6 kHz**.
2. (a) Nyquist rate = $2F_{max} = 2(6\text{ kHz}) =$ **12 kHz**.
3. (b) $F_s = 5$ kHz → folding frequency 2.5 kHz. $f_1 = 1/5$ (fine). $f_2 = 3/5 \to 3/5 - 1 =$ **$-2/5$**. $f_3 = 6/5 \to 6/5 - 1 =$ **$1/5$**.
4. $x(n) = 3\cos(2\pi n/5) + 5\sin(2\pi(-2/5)n) + 10\cos(2\pi n/5) =$ **$13\cos(2\pi n/5) - 5\sin(4\pi n/5)$**. (The sine picked up a minus sign because sin is odd; the two cosines at $f = 1/5$ add: 3 + 10.)
5. (c) Ideal D/A outputs frequencies inside $\pm F_s/2$: $f = 1/5 \to 1000$ Hz, $f = 2/5 \to 2000$ Hz. **$y_a(t) = 13\cos(2000\pi t) - 5\sin(4000\pi t)$**.

Sanity check: evaluate both expressions at $n = 1$ ($t = 0.2$ ms) — they agree to the last digit.

L1-04 · Periodicity of CT and DT sinusoids

Decide whether each signal is periodic and give the fundamental period. (a) $x_a(t) = 3\cos(5t + \pi/6)$ (b) $x(n) = \cos(0.01\pi n)$ (c) $x(n) = 3\cos(5n + \pi/6)$ (c') $x(n) = 3\cos(5\pi n + \pi/6)$ (d) $x(n) = \cos(\pi n/8)\cdot\cos(\pi n/3)$ (e) Are $\cos(5\pi n/2)$ and $\cos(\pi n/2)$ distinct or identical?

Signal	$f = \omega/2\pi$	Verdict
(a) $3\cos(5t+\pi/6)$	–	**CT sinusoids are always periodic: $T = 2\pi/5$ *s***
(b) $\cos(0.01\pi n)$	$0.005 = 1/200$	**Rational → periodic, $N = 200$**
(c) $3\cos(5n+\pi/6)$	$5/2\pi$	**Irrational → not periodic**
(c') $3\cos(5\pi n+\pi/6)$	5/2	**Rational → periodic, $N = 2$**
(d) $\cos(\pi n/8)\cos(\pi n/3)$	1/16 and 1/6	**$N_1 = 16$, $N_2 = 6 \to N = \text{lcm}(16, 6) = 48$**
(e) $\cos(5\pi n/2)$ vs $\cos(\pi n/2)$	5/4 and 1/4	**$5\pi/2 = 2\pi + \pi/2 \to$ identical (frequencies differ by an integer multiple of 2π)**

⚠ A DT sinusoid is periodic only if $f = k/N$ is a *ratio of integers*. Without a π in the argument (case c) $\omega/2\pi$ is irrational. Distinct DT sinusoids only exist for $|f| \le ½$ ($-\pi \le \omega \le \pi$); the fastest oscillation is $\omega = \pi$ (f = ½).

The slide shows these without the π symbols in the text layer; (b), (c) and (d) above assume the π is present. Case (c') is the contrast that makes the point.

L1-05 · Quantizer bits and SQNR (rounding vs truncation)

The DT signal $x(n) = 6.35\cos(\pi n/10)$ is quantized with resolution (a) $\Delta = 0.1$ and (b) $\Delta = 0.02$. How many bits does the A/D converter need in each case? Compute $SQNR_{dB}$ for each using rounding and using truncation.

1. Range: $x_{max} = 6.35$, $x_{min} = -6.35 \to 12.7$. Levels: $L = (x_{max} - x_{min})/\Delta + 1$.
2. (a) $L = 12.7/0.1 + 1 = 128 \Rightarrow b = \log_2 128 =$ **7 bits**.
3. (b) $L = 12.7/0.02 + 1 = 636 \Rightarrow 2^9 = 512 < 636 \le 1024 \Rightarrow$ **10 bits**.
4. Signal power of a cosine: $P_x = A^2/2 = 6.35^2/2 = 20.16$.
5. **Rounding:** error uniform on $(-\Delta/2, \Delta/2) \to P_q = \Delta^2/12$.
6. **Truncation:** error uniform on [0, Δ) → mean-square error $P_q = \Delta^2/3$ (what the MATLAB demo computes as mean(e²)); its *variance* is still $\Delta^2/12$ but the mean $\Delta/2$ is a DC bias.

Δ	Bits	SQNR round	SQNR trunc (mean-square)
0.1	7	**$10\log(20.16 / (0.01/12)) \approx 43.8$ dB**	$10\log(20.16 / (0.01/3)) \approx$ **37.8 dB**
0.02	10	**$10\log(20.16 / (0.0004/12)) \approx 57.8$ dB**	$10\log(20.16 / (0.0004/3)) \approx$ **51.8 dB**

Truncation is $10\log(4) = 6.02$ dB worse than rounding. Going from $\Delta = 0.1$ to 0.02 shrinks Δ by 5×, so SQNR rises by $20\log(5) = 13.98$ dB (57.8 − 43.8 ✓). Bits are rounded *up* (636 levels need 10 bits but only use 636 of 1024 codes), so

this is less than a full $3 \times 6.02 = 18$ dB. A MATLAB run on a few samples lands within about ±2 dB of these theoretical values.

L1-08 · Reading the A/D chain from a sample table

A 3-bit uniform quantizer covers −1 V … +1 V ($x_{min} = -1$, $x_{max} = +1$) using rounding. Find the number of levels, the step Δ, and the codeword and quantization error for the sample $x = 0.30$ V.

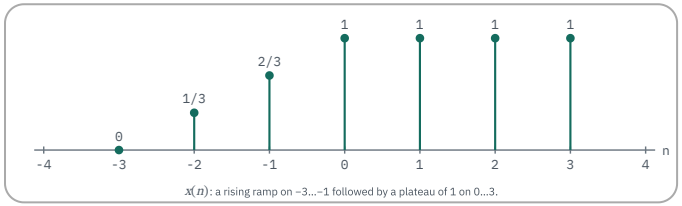
1. $L = 2^3 =$ **8 levels**.
2. $\Delta = (x_{max} - x_{min})/(L-1) = 2/7 =$ **0.2857 V** (the slide's s convention: levels include both end points).
3. Level k ($k = 0 \ldots 7$) sits at $-1 + k\cdot(2/7)$: −1, −0.714, −0.429, −0.143, +0.143, +0.429, +0.714, +1. The codeword is k in binary (000 … 111).
4. $x = 0.30$ lies between level 4 (+0.143) and level 5 (+0.429). Distances: 0.157 and 0.129 ⇒ rounds up to level 5: $x_q =$ **0.429 V**, **code 101**.
5. Quantization error $e_q = x_q - x = 0.429 - 0.300 =$ **+0.129 V**, and $|e_q| \le \Delta/2 = 0.143$ ✓.

⚠ Two conventions exist: Δ = range/$(L-1)$ (end points are levels, used on the slides) and Δ = range/L (midrise). Always say which one you used. The error sign convention also varies ($x - x_q$ or $x_q - x$).

L2-01 · Fold-then-delay vs delay-then-fold

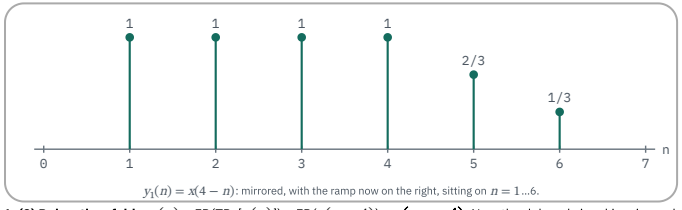
$x(n) = 1 + n/3$ for $-3 \le n \le -1$; $x(n) = 1$ for $0 \le n \le 3$; $x(n) = 0$ elsewhere. (a) Determine the values of $x(n)$. (b) Sketch (1) x folded and then delayed by four samples, (2) x delayed by four samples and then folded.

n	−4	−3	−2	−1	0	1	2	3	4
$x(n)$	0	0	1/3	2/3	1	1	1	1	0



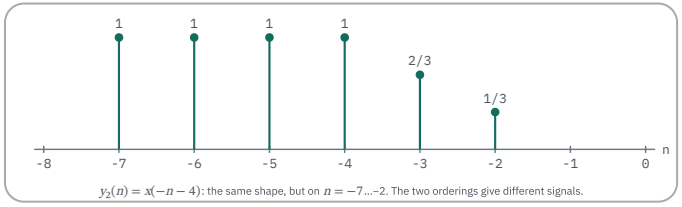
$x(n)$: a rising ramp on −3…−1 followed by a plateau of 1 on 0…3.

1. **(1) Fold, then delay:** $y_1(n) = \mathrm{TD}_4(\mathrm{FD}[x(n)]) = \mathrm{TD}_4(x(-n)) =$ **$x(-[n-4]) = x(4-n)$**. Fold is applied to n first, so the delay moves the *folded* signal right.
2. $x(4-n)$ is nonzero when $-3 \le 4-n \le 3 \Rightarrow 1 \le n \le 7$, with $n = 7$ giving $x(-3) = 0$. Values: $y_1(1 \ldots 6) = x(3), x(2), x(1), x(0), x(-1), x(-2) =$ **1, 1, 1, 1, 2/3, 1/3**.



$y_1(n) = x(4-n)$: mirrored, with the ramp now on the right, sitting on $n = 1 \ldots 6$.

1. **(2) Delay, then fold:** $y_2(n) = \mathrm{FD}(\mathrm{TD}_4[x(n)]) = \mathrm{FD}(x(n-4)) =$ **$x(-n-4)$**. Now the delayed signal is mirrored about $n = 0$, so it ends up on the *negative* side.
2. $x(-n-4)$ is nonzero for $-3 \le -n-4 \le 3 \Rightarrow -7 \le n \le -1$. Values: $y_2(-7 \ldots -2) = x(3), x(2), x(1), x(0), x(-1), x(-2) =$ **1, 1, 1, 1, 2/3, 1/3**.



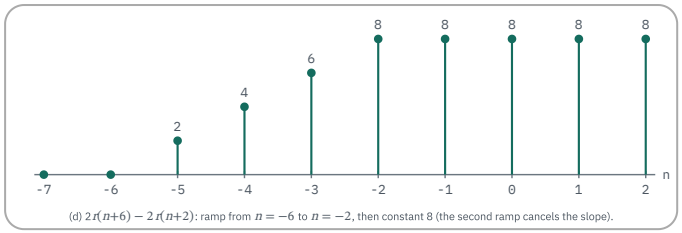
$y_2(n) = x(-n-4)$: the same shape, but on $n = -7 \ldots -2$. The two orderings give different signals.

⚠ Order matters. The trick: write the final formula with n replaced everywhere in $x(\cdot)$. Fold-then-delay replaces n by $-(n-4)$; delay-then-fold replaces n by $-n-4$.

L2-02 · Sketch standard signals; build $u(n)$ from impulses

Sketch and label (a) $x(n) = -2u(-n-4)$ (b) $x(n) = u(n+3) - u(n-4)$ (c) $x(n) = 8\delta(n-6)$ (d) $x(n) = 2r(n+6) - 2r(n+2)$. Then express the unit step as a function of impulses.

1. **(a)** $u(-n-4) = 1$ when $-n-4 \ge 0$, *i.e.* $n \le -4$. So $x(n) =$ **-2 for every $n \le -4$**, 0 for $n > -4$ (a left-sided step of height −2).
2. **(b)** $u(n+3)$ turns on at $n = -3$; $u(n-4)$ turns on at $n = 4$. Their difference is **1 for $-3 \le n \le 3$** (seven samples), 0 elsewhere — a rectangular window.
3. **(c)** A single sample of height **8 at $n = 6$**.
4. **(d)** $r(k) = k$ for $k \ge 0$. First term $2(n+6)$ starts at $n = -6$; second $-2(n+2)$ starts at $n = -2$. For $-6 \le n \le -2$ the ramp climbs 0, 2, 4, 6, 8; for $n \ge -2$ the slopes cancel and the output stays at **8**.



(d) $2r(n+6) - 2r(n+2)$: ramp from $n = -6$ to $n = -2$, then constant 8 (the second ramp cancels the slope).

$$u(n) = \sum_{k=0}^{\infty} \delta(n-k) \quad \text{(and conversely } \delta(n) = u(n) - u(n-1)\text{)}$$

Also $r(n) = \sum_{k} k\cdot\delta(n-k) = n\cdot u(n)$, and $u(n) - u(n-1) = \delta(n)$: each "derivative" of the chain ramp → step → impulse is a first difference.

L2-03 · Classify: $y(n) = n\cdot x(n-1)$, $y(n) = x(n^2+1)$, $y(n) = \cos x(n)$

Determine whether each system is static/dynamic, linear/non-linear, causal/non-causal and time-invariant/time-variant. Also decide BIBO stability.

Shift test for time-invariance: compute the response to $x(n-k)$ and compare with the delayed output $y(n-k)$.

1. **$y(n) = n\cdot x(n-1)$** — depends on $x(n-1)$: **dynamic**, past only ⇒ **causal**. Linear: $n\cdot[a\,x_1 + b\,x_2](n-1) = a\cdot y_1 + b\cdot y_2$ ⇒ **linear**. Time test: response to $x(n-k)$ is $n\cdot x(n-k-1)$; delayed output is $(n-k)\cdot x(n-k-1)$ — differ by the factor n vs $n-k$ ⇒ **time-variant**. Stability: $x = 1$ (bounded) gives $y = n$ (unbounded) ⇒ **unstable**.
2. **$y(n) = x(n^2+1)$** — dynamic. Causal needs $n^2 + 1 \le n$, but $n^2 - n + 1 > 0$ for every n, so the index is always in the *future* ⇒ **non-causal**. Linear ✓. Time test: response to $x(n-k)$ is $x(n^2 + 1 - k)$; $y(n-k) = x((n-k)^2 + 1)$ — differ ⇒ **time-variant**. Stable ($|y| \le \max|x|$).
3. **$y(n) = \cos x(n)$** — only the present input ⇒ **static**, **causal**. Not linear: $\cos(a\cdot x) \ne a\cdot\cos x$, and $\cos 0 = 1 \ne 0$ ⇒ **non-linear**. Time test: $\cos x(n-k) = y(n-k)$ ⇒ **time-invariant**. Stable ($|y| \le 1$).

Machine check: all three rows above were confirmed by testing shifted and superposed random inputs numerically (`verify.js`).

L2-04 · Quick-fire classification table (the slide traps)

Without long proofs, fill in static?, linear?, time-invariant?, causal?, BIBO-stable? for: (1) $y = 2x(n) + 3$ (2) $y = x^2(n)$ (3) $y = x(4n)$ (4) $y = x(n+3)$ (5) $y = 0.22x(n) + 0.1x(n-1)$ (6) $y = 0.22x(n) + 0.1x(n+1)$

System	Memory	Linear	Time-inv.	Causal	Stable	Key reason
$2x(n)+3$	static	**no**	yes	yes	yes	$x=0 \Rightarrow y=3 \ne 0$ (affine, fails homogeneity)
$x^2(n)$	static	**no**	yes	yes	yes	squares break additivity and homogeneity
$x(4n)$	dynamic	yes	**no**	**no**	yes	scaling n (down-sampling) is time-variant; $n=1$ needs $x(4)$
$x(n+3)$	dynamic	yes	yes	**no**	yes	pure advance by 3: needs future input
$0.22x(n)+0.1x(n-1)$	dynamic	yes	yes	yes	yes	FIR, present + past
$0.22x(n)+0.1x(n+1)$	dynamic	yes	yes	**no**	yes	impossible in real time (online)

$$\text{Linear} \Leftrightarrow T[a_1x_1 + a_2x_2] = a_1T[x_1] + a_2T[x_2]$$

⚠ "Static" means the output depends on the *present* input only. An easy necessary test for linearity: zero in ⇒ zero out. Any constant added to the output ($y = 2x + 3$) kills linearity, but time-invariance survives.

L2-05 · Convolution of finite sequences (fold-shift-multiply-sum)

Convolve: (a) $x(n) = \{2, -1, 0, 1\}$, $h(n) = \{1, 1, 0, -1\}$ (b) $x(n) = \{2, 0, 1\}$, $h(n) = \{1, 0, -1\}$. (All sequences start at $n = 0$.)

1. Output length = $L_x + L_h - 1 = 4 + 4 - 1 =$ **7** ($n = 0 \ldots 6$). Use $y(n) = \sum_k x(k)\,h(n-k)$.
2. $y(0) = 2\cdot 1 =$ **2**
3. $y(1) = 2\cdot 1 + (-1)\cdot 1 =$ **1**
4. $y(2) = 2\cdot 0 + (-1)\cdot 1 + 0\cdot 1 =$ **−1**
5. $y(3) = 2\cdot(-1) + (-1)\cdot 0 + 0\cdot 1 + 1\cdot 1 =$ **−1**
6. $y(4) = (-1)\cdot(-1) + 0\cdot 0 + 1\cdot 1 =$ **2**
7. $y(5) = 0\cdot(-1) + 1\cdot 0 =$ **0**
8. $y(6) = 1\cdot(-1) =$ **−1**

$$\text{(a)}\quad y(n) = \{2, 1, -1, -1, 2, 0, -1\}$$

Check: $\sum y = \sum x\cdot\sum h = (2)(1) = 2 + 1 - 1 - 1 + 2 + 0 - 1$ ✓.

(b) length $3 + 3 - 1 = 5$: $y(0) = 2$, $y(1) = 0$, $y(2) = 2\cdot(-1) + 0 + 1\cdot 1 = -1$, $y(3) = 0\cdot(-1) + 1\cdot 0 = 0$, $y(4) = 1\cdot(-1) = -1$.

$$\text{(b)}\quad y(n) = \{2, 0, -1, 0, -1\}$$

Check: $\sum x\cdot\sum h = (3)(0) = 0 = 2 + 0 - 1 + 0 - 1$ ✓.

Index of the first output = (first index of x) + (first index of h). If x starts at $n = -1$, so does the whole y shift by −1.

L2-06 · Convolution of two infinite causal exponentials

Determine the response of the system with $h(n) = (¼)^n u(n)$ to the input $x(n) = (½)^n u(n)$. Then generalize to $h = \alpha^n u(n)$, $x = \beta^n u(n)$.

1. Both causal ⇒ $y(n) = \sum_{k=0}^{n} x(k)\,h(n-k)$ for $n \ge 0$ (and 0 for $n < 0$).
2. $y(n) = \sum_{k=0}^{n} (½)^k(¼)^{n-k} = (¼)^n \sum_{k=0}^{n} 2^k = (¼)^n(2^{n+1} - 1)$.
3. Simplify: $(¼)^n\cdot 2^{n+1} = 2(½)^n$, so **$y(n) = [2(½)^n - (¼)^n]\,u(n)$**.
4. Check: $y(0) = 2 - 1 = 1 = x(0)\,h(0)$ ✓; $y(1) = 1 - ¼ = ¾ = x(0)\,h(1) + x(1)\,h(0) = ¼ + ½$ ✓.

$$\alpha^n u(n) * \beta^n u(n) = \frac{\alpha^{n+1} - \beta^{n+1}}{\alpha - \beta}\,u(n), \quad \alpha \ne \beta$$

⚠ If $\alpha = \beta$ the formula degenerates: $\alpha^n u(n) * \alpha^n u(n) = (n+1)\,\alpha^n u(n)$. The factor $(n+1)$ is the "repeated pole" signature you will meet again in the Z-transform.

L2-07 · Interconnected systems: find the equivalent $h(n)$

Systems: $h_1 = \delta(n) + \delta(n-1)$, $h_2 = \delta(n) - \delta(n-1)$, $h_3(n) = -\delta(n-1) + \delta(n-2)$, $h_4(n) = \delta(n-1)$. (a) Cascade of h_1 and h_2. (b) Parallel of h_1 and h_2. (c) h_1 in series with the parallel pair ($h_2 \parallel h_3$), all of that in parallel with h_4. (d) What is the cascade of $h(n) = (½)^n u(n)$ with $g(n) = \delta(n) - ½\delta(n-1)$?

1. **Rules:** cascade ⇒ $h = h_1 * h_2$ (order irrelevant, commutative); parallel ⇒ $h = h_1 + h_2$.
2. (a) $\{1, 1\} * \{1, -1\} =$ **$\{1, 0, -1\}$**, *i.e.* $\delta(n) - \delta(n-2)$.
3. (b) $h_1 + h_2 =$ **$2\delta(n)$**.
4. (c) Parallel pair first, over $n = 0 \ldots 2$: $h_2 = \{1, -1, 0\}$, $h_3 = \{0, -1, 1\}$ ⇒ $h_2 + h_3 = \{1, -2, 1\}$. Cascade with h_1: $\{1, 1\} * \{1, -2, 1\} = \{1, -1, -1, 1\}$. Add the parallel branch $h_4 = \delta(n-1) = \{0, 1, 0, 0\}$: **$h = \{1, 0, -1, 1\}$** ($n = 0 \ldots 3$).
5. (d) $(½)^n u(n) - ½(½)^{n-1} u(n-1) = (½)^n[u(n) - u(n-1)] =$ **$\delta(n)$**. The two systems are **inverses** — the cascade is the identity.

Method: do the innermost parallel sums first, then cascade (convolve), then add the outermost parallel branch. Write every impulse response as a list over a common index range before adding.

L2-10 · Using $h(n)$ with linearity + time-invariance

An LTI system has impulse response $h(n) = \delta(n) - \delta(n-1)$. Without convolving numerically, find its response to $x(n) = -\delta(n+1) + 2\delta(n) + 3\delta(n-2)$.

1. Linearity: scaling an impulse scales the response. Time-invariance: delaying an impulse delays the response. So $\delta(n-m) \to h(n-m)$.
2. $y(n) = -h(n+1) + 2\,h(n) + 3\,h(n-2)$.
3. $= -[\delta(n+1) - \delta(n)] + 2[\delta(n) - \delta(n-1)] + 3[\delta(n-2) - \delta(n-3)]$
4. $= -\delta(n+1) + \delta(n) + 2\delta(n) - 2\delta(n-1) + 3\delta(n-2) - 3\delta(n-3)$.

$$y(n) = -\delta(n+1) + 3\delta(n) - 2\delta(n-1) + 3\delta(n-2) - 3\delta(n-3)$$

This is *exactly* the convolution sum $y(n) = \sum x(k)\,h(n-k)$ written out term by term: each input sample launches a scaled, delayed copy of $h(n)$.

Equivalent list form: $x = \{-1, 2, 0, 3\}$ starting at $n = -1$, $h = \{1, -1\}$ ⇒ $y = \{-1, 3, -2, 3, -3\}$ starting at $n = -1$.

L3-01 · First-order LCCDE, full direct method

$y(n) - 0.5y(n-1) = x(n)$, $n \ge 0$, with $y(-1) = 1$. Obtain: (a) the homogeneous solution, (b) the zero-input response, (c) the particular solution for $x(n) = u(n)$, (d) the zero-state response for $x(n) = u(n)$, (e) the total response, (f) the impulse response $h(n)$. Identify the transient and steady-state parts.

1. **(a) Homogeneous.** Set $x = 0$, try $y_h = \lambda^n$: $\lambda^n - 0.5\,\lambda^{n-1} = 0 \Rightarrow \lambda = 0.5 \Rightarrow$ **$y_h(n) = C(0.5)^n$**.
2. **(b) Zero-input response** ($x = 0$, uses $y(-1) = 1$): $y(0) = 0.5\,y(-1) = 0.5 = C \Rightarrow$ **$y_{zi}(n) = 0.5\cdot(0.5)^n = (0.5)^{n+1}$**.
3. **(c) Particular** for $u(n)$: try $y_p = K$: $K - 0.5\,K = 1 \Rightarrow$ **$y_p = 2$**.
4. **(d) Zero-state** (relaxed, $y(-1) = 0$): $y = C(0.5)^n + 2$ and $y(0) = x(0) + 0.5\cdot 0 = 1 \Rightarrow C = -1 \Rightarrow$ **$y_{zs}(n) = 2 - (0.5)^n$**.
5. **(e) Total** = $y_{zi} + y_{zs} = 0.5(0.5)^n + 2 - (0.5)^n =$ **$2 - 0.5(0.5)^n$**. Check: $y(0) = 1 + 0.5(1) = 1.5 = 2 - 0.5$ ✓.
6. **(f) Impulse response**: ZSR to $\delta(n)$: the particular solution is zero for $n \ge 1$, so $y = C(0.5)^n$ with $y(0) = 1 \Rightarrow$ **$h(n) = (0.5)^n\,u(n)$**.

Transient = the decaying part, **$-0.5(0.5)^n$** (same form as y_h); steady state = the part shaped like the input, **2** (= y_p).

⚠ Two ways to get the total: add ZIR + ZSR (above) *or* solve $y_h + y_p$ once and fit C using $y(0)$ computed *with* the nonzero initial condition. Never fit C from $y(-1)$ alone after y_p is included — $y(-1)$ is not in the range $n \ge 0$ of the solution.

L3-04 · Recursing by hand; ZIR vs ZSR; $h(n)$ of a first-order system

$y(n) = 0.1y(n-1) + 0.2x(n) - 0.15x(n-1)$. (a) With a relaxed system, write $y(0)$, $y(1)$ in terms of the inputs. (b) Find the impulse response $h(n)$. (c) With $x = 0$ and $y(-1) = 1$, find the zero-input response.

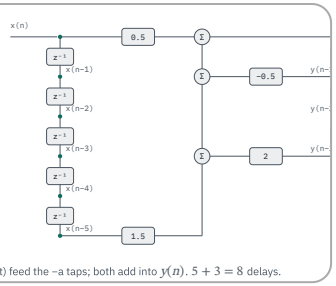
1. (a) Relaxed: $x(-1) = 0$, $y(-1) = 0$. $y(0) =$ **$0.2x(0)$**. $y(1) = 0.1\,y(0) + 0.2x(1) - 0.15x(0) = 0.02x(0) + 0.2x(1) - 0.15x(0) =$ **$0.2x(1) - 0.13x(0)$**.
2. (b) Feed $x = \delta(n)$: $h(0) = 0.2$; $h(1) = 0.1(0.2) - 0.15 =$ **−0.13**; for $n \ge 2$ the input terms vanish and $h(n) = 0.1\,h(n-1)$.
3. So **$h(n) = -0.13(0.1)^{n-1}$** for $n \ge 1$, with $h(0) = 0.2$. Equivalent compact form: $h(n) = 0.2(0.1)^n u(n) - 0.15(0.1)^{n-1}u(n-1)$ — each term is the output of the recursive part driven by one input tap.
4. (c) ZIR: $x = 0$, so $y(n) = 0.1\,y(n-1)$: $y(0) = 0.1$, $y(1) = 0.01$, … ⇒ **$y_{zi}(n) = (0.1)^{n+1}$**, $n \ge 0$.

A recursive system with one feedback tap has an infinite $h(n)$ (IIR); here it decays by a factor 0.1 per sample. The feedforward taps (0.2, −0.15) only decide the first two samples.

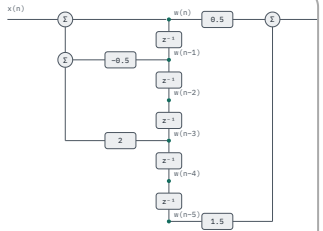
L3-05 · Direct Form I and II realization (and a stability surprise)

Sketch the Direct Form I and Direct Form II realizations of $2y(n) + y(n-1) - 4y(n-3) = x(n) + 3x(n-5)$. How many delay elements does each use? Is the system stable?

1. **Normalize first.** Divide by 2 so the coefficient of $y(n)$ is 1: $y(n) = -0.5\,y(n-1) + 2\,y(n-3) + 0.5\,x(n) + 1.5\,x(n-5)$.
2. So **$a_1 = 0.5$, $a_2 = 0$, $a_3 = -2$** and **$b_0 = 0.5$, $b_5 = 1.5$** ($b_1 \ldots b_4 = 0$). $N = 3$, $M = 5$.
3. **DF-I** keeps separate delay lines for x ($M = 5$ delays) and y ($N = 3$ delays): **8 delays**. The drawn feedback multipliers are **$-a_k$**: −0.5 and +2.
4. **DF-II** shares one line $w(n) = x(n) - a_1\,w(n-1) - \ldots$ and needs only $\max(M, N) =$ **5 delays**.



Direct Form I: x-delays (left) feed the b taps, y-delays (right) feed the −a taps; both add into $y(n)$. $5 + 3 = 8$ delays.



Direct Form II: one shared delay line $w(n-k)$ with feedback taps $-a_k$ on the left and feedforward taps b_k on the right. 5 delays.

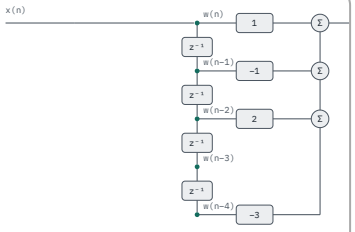
1. **Stability.** $H(z) = (0.5 + 1.5z^{-5})/(1 + 0.5z^{-1} - 2z^{-3})$. Poles solve $z^3 + 0.5z^2 - 2 = 0$: one real root at $z \approx$ **1.113** and a complex pair of magnitude ≈ **1.340**. All three lie **outside** the unit circle ⇒ the causal system is **unstable** (outputs grow without bound).

⚠ The numerator (b) never affects stability; only the denominator coefficients a_k do. Realizing a system does not make it good — check the poles before you build it.

L3-06 · FIR realization and impulse response

Obtain the direct-form realization of $y(n) = x(n) - x(n-1) + 2x(n-2) - 3x(n-4)$ and solve for its impulse response. Is it recursive? Stable?

1. Let $x = \delta(n)$: $y(n) = \delta(n) - \delta(n-1) + 2\,\delta(n-2) - 3\,\delta(n-4)$.
2. $h(n) = \{1, -1, 2, 0, -3\}$ for $n = 0 \ldots 4$ (note the zero tap at $n = 3$ – it is still part of the sequence).
3. No past outputs appear ⇒ **non-recursive**; finite duration ⇒ **FIR**. $H(z) = 1 - z^{-1} + 2z^{-2} - 3z^{-4}$ has all four poles at $z = 0$ ⇒ always **BIBO stable** ($\Sigma|h| = 7$).
4. Needs **4 delays** (highest delay is $x(n-4)$); the $n-3$ tap is simply left out.



FIR (moving-average / tapped-delay) realization: no feedback. The $b_3 = 0$ tap is omitted; the adder chain passes straight through that row.

L3-07 · Block diagram → difference equation

Find the difference equation of the system in the figure (Direct Form II; the multipliers are as drawn).

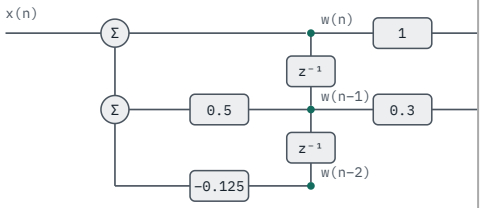


Figure for the problem. Drawn feedback multipliers: +0.5 on $w(n-1)$ and −0.125 on $w(n-2)$. Feedforward taps: 1 on $w(n)$, 0.3 on $w(n-1)$.

1. Name the internal node $w(n)$ at the top of the delay line. The feedback adder gives $w(n) = x(n) + 0.5\,w(n-1) - 0.125\,w(n-2)$ (drawn multipliers add as drawn).
2. The output adder gives $y(n) = w(n) + 0.3\,w(n-1)$.
3. Eliminate w: apply the same operator to y. Since $y(n) = w(n) + 0.3\,w(n-1)$ is just w filtered by a feedforward polynomial, y satisfies the same recursion on the left and the polynomial on the right:

$$y(n) = 0.5y(n-1) - 0.125y(n-2) + x(n) + 0.3x(n-1)$$

In the standard form $y + a_1 y(n-1) + a_2 y(n-2) = \ldots$, this means $a_1 = -0.5$, $a_2 = +0.125$. Poles: $z^2 - 0.5z + 0.125 = 0 \Rightarrow |z| = \sqrt{0.125} \approx 0.354$ (complex pair) ⇒ stable.

⚠ The drawn feedback gain is $-a_k$. When you read a diagram, write the equation exactly as drawn first, *then* convert to the a_k convention.

L4-01 · Z-transform and ROC of finite sequences

$x(n) = \{3, -2, -2, 5, 1, 4, -1\}$ where the arrow ($n = 0$) is on the "5". (a) Find $X(z)$ and its ROC. (b) What changes if $n = 0$ is the first sample? The last sample? (c) $X(z)$ and ROC of $\delta(n)$.

1. (a) Indices run $n = -3 \ldots 3$. $X(z) = \Sigma\, x(n) z^{-n}$ gives $X(z) = 3z^3 - 2z^2 - 2z + 5 + z^{-1} + 4z^{-2} - z^{-3}$.
2. It has positive powers (z^3) ⇒ blows up as $z \to \infty$; negative powers (z^{-3}) ⇒ blows up as $z \to 0$. **ROC: all z except 0 and ∞** ($0 < |z| < \infty$).
3. (b) $n = 0$ on the *first* sample (causal): $X = 3 - 2z^{-1} - 2z^{-2} + 5z^{-3} + z^{-4} + 4z^{-5} - z^{-6}$, only negative powers ⇒ **ROC: $|z| > 0$** (includes ∞). $n = 0$ on the *last* sample (anti-causal): $X = 3z^6 - 2z^5 - 2z^4 + 5z^3 + z^2 + 4z - 1$, only positive powers ⇒ **ROC: $|z| < \infty$** (includes 0).
4. (c) $\delta(n) \to X(z) = 1$, ROC **all z**.

Signal	Has terms	ROC
$x(n) \neq 0$ only for $n \geq 0$	z^{-k} only	**$\lvert z\rvert > 0$ (∞ included)**
$x(n) \neq 0$ only for $n \leq 0$	z^{k} only	**$\lvert z\rvert < \infty$ (0 included)**
both	z^{-k} and z^{k}	**$0 < \lvert z\rvert < \infty$**
$\delta(n)$	none (constant)	**all z**

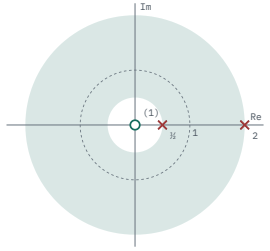
A finite-length sequence's s ROC is the whole plane minus whichever of 0, ∞ the powers of z force out. The ROC never contains a pole.

L4-02 · Infinite sequences: causal, anti-causal, two-sided, and no ROC

Find $X(z)$ and the ROC: (a) $x(n) = a^n u(n)$ (b) $x(n) = -a^n u(-n-1)$ (c) $x(n) = (½)^n u(n) + 2^n u(-n-1)$ (d) $x(n) = 2^n u(n) + (½)^n u(-n-1)$.

1. **(a)** $\Sigma_{n\geq 0}(az^{-1})^n = \mathbf{1/(1 - az^{-1})}$, converges for $|az^{-1}| < 1 \Rightarrow$ **ROC $|z| > |a|$** (outside the pole — right-sided).
2. **(b)** $-\Sigma_{n\leq -1}(az^{-1})^n = -\Sigma_{m\geq 1}(a^{-1}z)^m = \mathbf{1/(1 - az^{-1})}$ again, but converges for $|a^{-1}z| < 1 \Rightarrow$ **ROC $|z| < |a|$** (inside the pole — left-sided). Same $X(z)$, different ROC, *different signal.*
3. **(c)** Causal part: $1/(1 - ½z^{-1})$, $|z| > ½$. Anti-causal part $2^n\,u(-n-1) = -[-2^n\,u(-n-1)] \to -1/(1 - 2z^{-1})$, $|z| < 2$. The ROCs overlap in an **annulus $½ < |z| < 2$**:

$$X(z) = \frac{1}{1 - ½z^{-1}} - \frac{1}{1 - 2z^{-1}} = \frac{-1.5z^{-1}}{(1 - ½z^{-1})(1 - 2z^{-1})}$$



(c) Poles at ½ and 2, zero at the origin. The ROC (shaded) is the ring between the poles — it contains the unit circle, so this two-sided signal has a Fourier transform.

1. **(d)** Needs $|z| > 2$ *and* $|z| < ½$ — the regions do not overlap. **No ROC ⇒ the Z-transform does not exist** (the sum diverges for every z).

⚠ $X(z)$ alone does not determine $x(n)$; $X(z)$ **and** its ROC do. Right-sided → outside the outermost pole; left-sided → inside the innermost pole; two-sided → a ring between poles.

L4-03 · Properties workout (shift, scaling, differentiation, reversal, convolution, initial value)

Find the Z-transform (use the properties, not the definition): (a) $(½)^{n-2}u(n-2)$ and $(½)^n u(n-2)$ (b) $n(½)^n u(n)$ (c) $(-½)^n \cos(\pi n/3)\,u(n)$ (d) $u(-n)$ (e) $u(n) * u(n)$ (f) $x(0)$ for $X(z) = (2 + z^{-1})/(1 - ½z^{-1})$ (causal).

1. **(a) Shift:** $x(n-2) \leftrightarrow z^{-2}X(z)$. With $x(n) = (½)^n\,u(n)$: $\mathbf{(½)^{n-2}u(n-2) \leftrightarrow z^{-2}/(1 - ½z^{-1})}$. The second signal is NOT shifted consistently: $(½)^n u(n-2) = ¼\cdot(½)^{n-2}u(n-2) \leftrightarrow ¼\,z^{-2}/(1 - ½z^{-1})$.
2. **(b) Differentiation:** $Z\{n\cdot x(n)\} = -z\,\mathrm{d}X/\mathrm{d}z$. $X = (1 - ½z^{-1})^{-1} \Rightarrow \mathrm{d}X/\mathrm{d}z = -(1 - ½z^{-1})^{-2}\cdot(½z^{-2}) \Rightarrow X_b = ½z^{-1}/(1 - ½z^{-1})^2$, $|z| > ½$.
3. **(c) Scaling:** $Z\{a^n x(n)\} = X(a^{-1}z)$. Start with $\cos(\pi n/3)\,u(n) \leftrightarrow (1 - ½z^{-1})/(1 - z^{-1} + z^{-2})$, then replace z^{-1} by a z^{-1} with a = −½: $\mathbf{(1 + ¼z^{-1})/(1 + ½z^{-1} + ¼z^{-2})}$, $|z| > ½$.
4. **(d) Reversal:** $x(-n) \leftrightarrow X(z^{-1})$. $u(n) \leftrightarrow 1/(1 - z^{-1})$, so $\mathbf{u(-n) \leftrightarrow 1/(1 - z)}$, ROC $|z| < 1$.
5. **(e) Convolution:** product of transforms: $(1/(1 - z^{-1}))^2 = 1/(1 - z^{-1})^2 \leftrightarrow \mathbf{(n+1)u(n)}$, ROC $|z| > 1$.
6. **(f) Initial value** (causal x): $x(0) = \lim_{z\to\infty} X(z) = (2 + 0)/(1 - 0) = \mathbf{2}$. (Series: $(2 + w)(1 + ½w + \ldots)$ gives $x(1) = 1 + 1 = 2$.)

⚠ The shift property moves the *whole* expression, argument included. If a^n is left unshifted while $u(n)$ is shifted, you must rewrite it first, as in (a).

Check on (c): $(-½)^n\cos(\pi n/3) = (½)^n\cos(2\pi n/3)$ since $(-1)^n\cos(\pi n/3) = \cos(2\pi n/3)$. The standard form $(1 - r\cos\omega\,z^{-1})/(1 - 2r\cos\omega\,z^{-1} + r^2z^{-2})$ with $r = ½$, $\omega = 2\pi/3$ gives the same answer.

L4-04 · Inverse Z by power series (long division)

$X(z) = 1/(1 - 1.5z^{-1} + 0.5z^{-2})$. Find the first few terms of $x(n)$ if (a) ROC $|z| > 1$, (b) ROC $|z| < 0.5$.

1. **(a) $|z| > 1$ ⇒ right-sided:** divide in descending powers of z (i.e. powers of z^{-1}).
2. $1 \div (1 - 1.5z^{-1} + 0.5z^{-2})$: 1 → remainder $1.5z^{-1} - 0.5z^{-2} \to 1.5z^{-1}$ → remainder $1.75z^{-2}$ … giving $X = \mathbf{1 + 1.5z^{-1} + 1.75z^{-2} + 1.875z^{-3} + \ldots}$ so $x(n) = \{1, 1.5, 1.75, 1.875, \ldots\}$ for $n = 0, 1, 2, 3$.
3. Pattern (confirmed by partial fractions in L4-05): $x(n) = 2 - (0.5)^n$ for $n \geq 0$.
4. **(b) $|z| < 0.5$ ⇒ left-sided:** divide in *ascending* powers of z. Rewrite $X = 2z^2/(1 - 3z + 2z^2)$ and expand $1/(1 - 3z + 2z^2) = 1 + 3z + 7z^2 + 15z^3 + \ldots$ (each coefficient $c_k = 3c_{k-1} - 2c_{k-2}$).
5. $\mathbf{X = 2z^2 + 6z^3 + 14z^4 + 30z^5 + \ldots} \Rightarrow x(-2) = 2$, $x(-3) = 6$, $x(-4) = 14$, $x(-5) = 30$ and $x(n) = 0$ for $n > -2$. Closed form: $x(n) = [-2 + (0.5)^n]$ for $n \leq -1$ (gives 0 at $n = -1$ ✓).

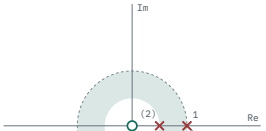
⚠ The ROC decides which way you divide: outside the outermost pole → descending powers of z (causal); inside the innermost pole → ascending powers of z (anti-causal). Power series is good for the first few samples; partial fractions give the closed form.

L4-05 · Inverse Z by partial fractions: distinct poles, three ROCs

$X(z) = 1/(1 - 1.5z^{-1} + 0.5z^{-2})$. Find the closed form of $x(n)$ for (a) $|z| > 1$, (b) ½ < $|z|$ < 1, (c) $|z| < ½$.

1. Factor the denominator: $1 - 1.5z^{-1} + 0.5z^{-2} = (1 - z^{-1})(1 - 0.5z^{-1})$, so the poles are $\mathbf{z = 1}$ **and** $\mathbf{z = 0.5}$.
2. $X(z) = A/(1 - z^{-1}) + B/(1 - 0.5z^{-1})$. Residues: A = $[(1 - z^{-1})X]_{z=1} = 1/(1 - 0.5) =$ **2**; $B = [(1 - 0.5z^{-1})X]_{z=0.5} = 1/(1 - 2) =$ **−1**.
3. $X(z) = 2/(1 - z^{-1}) - 1/(1 - 0.5z^{-1})$. Now assign each term's s side from the ROC:

ROC	pole 1 term	pole ½ term	$x(n)$
$\lvert z\rvert > 1$	right-sided: $2\,u(n)$	right-sided: $-(½)^n u(n)$	$\mathbf{2\,u(n) - (½)^n u(n)}$
½ < $\lvert z\rvert$ < 1	left-sided: $-2\,u(-n-1)$	right-sided: $-(½)^n u(n)$	$\mathbf{-2\,u(-n-1) - (½)^n u(n)}$
$\lvert z\rvert < ½$	left-sided: $-2\,u(-n-1)$	left-sided: $+(½)^n u(-n-1)$	$\mathbf{[-2 + (½)^n]\,u(-n-1)}$



Case (b): the ROC is the ring between the poles, so the pole at ½ is treated as causal and the pole at 1 as anti-causal. Two zeros sit at the origin.

Part (a) matches the long-division series in L4-04: $2 - (½)^n = 1, 1.5, 1.75, 1.875, \ldots$ ✓. In case (c) the $n = -1$ sample is 0 $(-2 + 2)$ and $n = -2$ gives 2 – also matching L4-04(b).

⚠ In case (b) the signal is two-sided and the unit circle is in the ROC only if the ROC contains $|z| = 1$. The ring ½ < $|z|$ < 1 does *not* contain it (the pole at $z = 1$ is on the boundary).

MATLAB check: `[r,p,k] = residuez(b,a)` expands in *negative* powers of z (b = numerator, a = denominator, in powers of z^{-1}; the form used here, and the denominator order must be higher than the numerator, i.e. proper). `residue` works in *positive* powers, so apply it to $X(z)/z$. For this X: b = 1, a = [1 −1.5 0.5] gives r = [2 −1], p = [1 0.5]. Same A and B as above (MATLAB may list the poles in a different order).

L4-06 · Improper rational $X(z)$: divide first

$X(z) = (1 + z^{-1} + z^{-2})/(1 - ½z^{-1})$, causal (ROC $|z| > ½$). Find $x(n)$.

1. Numerator degree in z^{-1} is 2, denominator degree is 1 ⇒ $M \geq N$ ⇒ **improper**. Divide polynomials in $w = z^{-1}$: $(w^2 + w + 1) \div (1 - ½w)$.
2. Quotient $-2w - 6$, remainder 7: check $(-2w - 6)(1 - ½w) + 7 = -2w + w^2 - 6 + 3w + 7 = w^2 + w + 1$ ✓.
3. So $\mathbf{X(z) = -6 - 2z^{-1} + 7/(1 - ½z^{-1})}$: a polynomial part (finite, just impulses) plus a proper fraction.
4. Invert term by term: $\mathbf{x(n) = -6\,\delta(n) - 2\,\delta(n-1) + 7(½)^n u(n)}$.
5. Check: $x(0) = -6 + 7 = 1$ ✓ (the leading numerator coefficient); $x(1) = -2 + 3.5 = 1.5$ and series $(1 + w + w^2)(1 + ½w + \ldots)$ gives $1 + ½ = 1.5$ ✓; $x(2) = 7(¼) = 1.75$ ✓.

Improper $X(z)$ always contributes impulses $\delta(n-k)$ at the start of $x(n)$. Skipping the division and trying $A/(1 - ½z^{-1})$ alone loses them.

L4-07 · Inverse Z with a repeated pole

$X(z) = z^{-1}/[(1 - z^{-1})(1 - ½z^{-1})^2]$, causal. Find $x(n)$.

1. Multiply top and bottom by z^2 and work with $X(z)/z$ so every factor has the form $z/(z - p)$: $X(z) = z^2/[(z-1)(z-½)^2] \Rightarrow \mathbf{X(z)/z = z/[(z-1)(z-½)^2]}$.
2. Expand: $X(z)/z = A/(z-1) + B/(z-½) + C/(z-½)^2$.
3. A = $[z/(z-½)^2]_{z=1} = 1/¼ =$ **4**. $C = [z/(z-1)]_{z=½} = ½/(-½) =$ **−1**. For the repeated root: $B = \mathrm{d}/\mathrm{d}z[z/(z-1)]_{z=½} = [-1/(z-1)^2]_{z=½} = -1/¼ =$ **−4**. (Check: $X(z)/z \sim 1/z^2$ at infinity ⇒ $A + B = 0$ ✓.)
4. $X(z) = 4\,z/(z-1) - 4\,z/(z-½) - z/(z-½)^2$. Using $z/(z-p)^2 \leftrightarrow n\,p^{n-1}\,u(n)$ (differentiation property): the last term gives $-n(½)^{n-1} = -2n(½)^n$.

$$x(n) = [4 - 4(½)^n - 2n(½)^n]\,u(n)$$

Check: $x(0) = 4 - 4 = 0$ ✓ (numerator has the extra z^{-1}); $x(1) = 4 - 2 - 1 = 1$ ✓; $x(2) = 4 - 1 - 1 = 2$ ✓ (series: $z^{-1}(1 + 2z^{-1} + \ldots) \to 0, 1, 2, \ldots$).

Each extra repetition of a pole adds a factor of n (n, n^2, …): a double pole gives $(C_1 + C_2 n)p^n$ — the same pattern you saw for repeated characteristic roots in Lec 3.

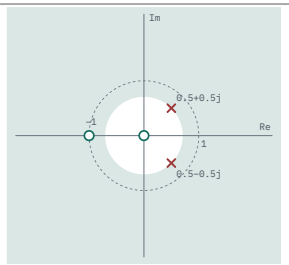
L4-08 · Inverse Z with complex-conjugate poles

$X(z) = (1 + z^{-1})/(1 - z^{-1} + 0.5z^{-2})$, causal. Find $x(n)$ as a real signal.

1. Denominator zeros: $z^2 - z + 0.5 = 0 \Rightarrow \mathbf{p = 0.5 \pm j0.5 = (1/\sqrt{2})e^{\pm j\pi/4}}$, $|p| = 0.707 < 1$.
2. $X(z)/z = (z + 1)/[(z-p)(z-p^*)]$. A = $[(z+1)/(z-p^*)]_{z=p} = (p + 1)/(p - p^*) = (1.5 + 0.5\mathrm{j})/(\mathrm{j}1) =$ **0.5 − 1.5j**. The partner residue is A*.
3. $x(n) = A\,p^n + A^*\,p^{*n} = 2|A|\,|p|^n\cos(n\theta + \angle A)\;u(n)$, with |A| = √(0.25 + 2.25) = √2.5 = 1.581, ∠A = atan2(−1.5, 0.5) = −71.57° = −1.249 rad, $\theta = \pi/4$.

$$x(n) = \sqrt{10}\,(1/\sqrt{2})^n\cos(\pi n/4 - 1.249)\,u(n)$$

Check: $x(0) = 3.162\cdot\cos(-71.57°) = 3.162\cdot0.316 = 1$ ✓; $x(1) = 3.162\cdot0.707\cdot\cos(45° - 71.57°) = 2.236\cdot0.894 = 2$ ✓; long division gives $x = \{1, 2, 1.5, 0.5, -0.25, \ldots\}$ ✓.



Poles at radius 0.707 and angle ±45° ⇒ a decaying oscillation with period 8 samples. Zeros at 0 and −1. ROC $|z| > 0.707$ contains the unit circle: stable.

⚠ Conjugate poles always give conjugate residues, so the imaginary parts cancel and $x(n)$ is real. Pole radius sets the decay rate ($|p|^n$), pole angle sets the oscillation frequency (θ rad/sample).

L4-09 · Convolution through the Z-transform

Use Z-transforms to convolve $x(n) = (½)^n u(n)$ with $h(n) = u(n)$.

1. $X(z) = 1/(1 - ½z^{-1})$, $H(z) = 1/(1 - z^{-1})$. Convolution in time ⇒ product in z: $\mathbf{Y(z) = 1/[(1 - ½z^{-1})(1 - z^{-1})]}$, ROC $|z| > 1$.
2. Partial fractions (same Y as L4-05): $A = 2$ at $z = 1$, $B = -1$ at $z = ½$.
3. $\mathbf{y(n) = [2 - (½)^n]\,u(n)}$.
4. Direct check: $\Sigma_{k=0}^{n}(½)^k = 2(1 - (½)^{n+1}) = 2 - (½)^n$ ✓.

The ROC of a product is at least the overlap of the two ROCs ($|z| > ½$ and $|z| > 1 \Rightarrow |z| > 1$). Pole–zero cancellation can enlarge it.

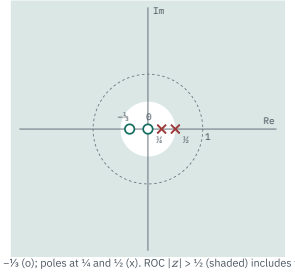
L4-10 · System function, poles/zeros, $h(n)$, step response

$y(n) - (3/4)y(n-1) + (1/8)y(n-2) = x(n) + (1/3)x(n-1)$, causal and relaxed. Find $H(z)$, the poles and zeros, $h(n)$, the stability verdict, and the unit-step response.

1. Take the Z-transform of both sides (zero initial conditions) and divide: $\mathbf{H(z) = (1 + ⅓z^{-1})/(1 - ¾z^{-1} + ⅛z^{-2})} = z(z + ⅓)/[(z - ½)(z - ¼)]$.
2. **Zeros:** $z = 0$ and $z = -⅓$. **Poles:** $z = ½$ and $z = ¼$ (solve $z^2 - ¾z + ⅛ = 0$).
3. **Stability:** causal ⇒ ROC is $|z| > ½$, which contains the unit circle; both poles inside ⇒ **BIBO stable**.
4. $\mathbf{h(n)}$**:** $H(z)/z = (z + ⅓)/[(z-½)(z-¼)]$ ⇒ residue at ½: $(½ + ⅓)/(¼) = 10/3$; at ¼: $(¼ + ⅓)/(-¼) = -7/3$. So $\mathbf{h(n) = [(10/3)(½)^n - (7/3)(¼)^n]\,u(n)}$. Check $h(0) = 1$ ✓, $h(1) = ¾ + ⅓ = 13/12$ ✓.
5. **Step response:** $Y(z) = H(z)\cdot1/(1 - z^{-1})$, so $Y(z)/z = z(z + ⅓)/[(z-1)(z-½)(z-¼)]$ with residues 32/9 (z=1), −10/3 (z=½), 7/9 (z=¼).

$$y_{step}(n) = [\frac{32}{9} - \frac{10}{3}(½)^n + \frac{7}{9}(¼)^n]\ u(n)$$

Steady state $32/9 = 3.556 = H(1) = (4/3)/(1 - ¾ + ⅛)$ ✓ (DC gain = $H(z)$ evaluated at $z = 1$). Check $y(0) = 32/9 - 30/9 + 7/9 = 1$ ✓.



Zeros at 0 and −⅓ (o); poles at ¼ and ½ (x). ROC $|z| > ½$ (shaded) includes the unit circle.

L4-11 · One-sided Z-transform with nonzero initial conditions

$y(n) = 0.9y(n-1) - 0.81y(n-2) + x(n)$ with $y(-1) = 1$, $y(-2) = 1$. Solve for the zero-input response.

1. One-sided shifts: $Z^+\{y(n-1)\} = z^{-1}Y(z) + y(-1)$; $Z^+\{y(n-2)\} = z^{-2}Y(z) + z^{-1}y(-1) + y(-2)$.
2. Set $x = 0$: $Y(z) = 0.9[z^{-1}Y + y(-1)] - 0.81[z^{-2}Y + z^{-1}y(-1) + y(-2)]$.
3. Collect: $Y(z)(1 - 0.9z^{-1} + 0.81z^{-2}) = 0.9y(-1) - 0.81y(-2) - 0.81y(-1)z^{-1} = \mathbf{0.09 - 0.81z^{-1}}$.
4. Poles: $1 - 0.9z^{-1} + 0.81z^{-2} \Rightarrow 2r\cos\theta = 0.9$, $r^2 = 0.81 \Rightarrow \mathbf{r = 0.9}$, $\mathbf{\theta = 60° = \pi/3}$ (complex pair 0.9∠±60°).
5. For $N(z) = n_0 + n_1z^{-1}$ over $1 - 2r\cos\theta\, z^{-1} + r^2z^{-2}$: $\mathbf{y(n) = r^n[n_0\cos n\theta + ((n_1 + n_0\, r\cos\theta)/(r\sin\theta))\sin n\theta]}$. Here $n_0 = 0.09$, $n_1 = -0.81$, $r\cos\theta = 0.45$, $r\sin\theta = 0.7794$: coefficient of sin = $(-0.81 + 0.0405)/0.7794 = -0.987 = -0.57\sqrt{3}$.

$$y_{zi}(n) = (0.9)^n[0.09\cos(\pi n/3) - 0.987\sin(\pi n/3)]\ u(n)$$

Check by recursion: $y(0) = 0.9(1) - 0.81(1) = 0.09$ ✓; $y(1) = 0.9(0.09) - 0.81(1) = -0.729$ ✓ (formula: $0.9[0.045 - 0.855] = -0.729$ ✓).

Total response = y_{zi} (from the initial conditions, denominator-only dynamics) + y_{zs} (from $H(z)X(z)$). The one-sided shift property is the only property that differs from the two-sided transform; it is exactly what injects the initial conditions.

L4-12 · Same $H(z)$, different ROC: causal vs stable

$H(z) = 1/(1 - 2z^{-1})$. Find $h(n)$ for each ROC: (a) $|z| > 2$, (b) $|z| < 2$. Which is causal? Which is stable? Can one choice be both?

1. The only pole is $z = 2$.
2. **(a) $|z| > 2$:** right-sided ⇒ $\mathbf{h(n) = 2^n\,u(n)}$. **Causal**, but $\Sigma|h| = \infty$ and the unit circle is *not* in the ROC ⇒ **unstable**.
3. **(b) $|z| < 2$:** left-sided ⇒ $\mathbf{h(n) = -2^n\,u(-n-1)}$. $\Sigma_{n\le-1}\,2^n = \Sigma_{k\ge1}\,2^{-k} = 1$ ⇒ **stable** (unit circle inside the ROC), but **non-causal** (nonzero for negative n).
4. No ROC gives both: causal needs the ROC outside the outermost pole, stable needs the unit circle in the ROC, so a causal stable system needs **every pole inside the unit circle**. This pole (2) fails.

Property	Condition on ROC
Causal	exterior of a circle, $\|z\| > r_{max}$ (and includes ∞)
Anti-causal	interior of a circle, $\|z\| < r_{min}$
BIBO stable	contains the unit circle $\|z\| = 1$
Causal and stable	all poles satisfy $\|p\| < 1$

Stability is never a property of $H(z)$ alone: it needs the ROC (or, for a causal system, the pole locations).

L4-13 · Pole location → shape of the impulse response

For a causal system describe the long-term behavior of $h(n)$ when the (only) poles are: (a) 0.9 (b) −0.9 (c) $0.9e^{\pm j\pi/4}$ (d) 1.1 (e) $e^{\pm j\pi/2}$ (f) 1 (g) a double pole at 0.9.

Poles	Term in $h(n)$	Behavior
(a) 0.9	$(0.9)^n$	**smooth monotone decay (slow: time constant ≈ $1/(-\ln 0.9) \approx 9.5$ samples)**
(b) −0.9	$(-0.9)^n$	**decays with sign alternating every sample**
(c) 0.9∠±45°	$(0.9)^n\cos(\pi n/4 + \varphi)$	**decaying oscillation, period $2\pi/(\pi/4) = 8$ samples**
(d) 1.1	$(1.1)^n$	**grows without bound — unstable**
(e) ∠±90°, $\|p\| = 1$	$\cos(\pi n/2 + \varphi)$	**sustained oscillation, period 4 — marginal (not BIBO stable): a digital oscillator**
(f) 1	constant	**does not decay ($h = u(n)$) — marginal; the accumulator**
(g) double at 0.9	$n(0.9)^n$	**rises, peaks, then decays: n·0.9^n peaks at $n = 9$ and 10 (equal, 3.487), then dies out**

$|p| < 1$: decays $|p| = 1$: constant/oscillating $|p| > 1$: grows

Radius → rate of decay or growth. Angle → frequency of oscillation (θ rad/sample ⇒ $F = \theta\cdot F_s/2\pi$). Negative real pole → angle π → fastest alternation. Multiplicity m → polynomial factor n^{m-1}.

L4-14 · Z-transforms of sinusoids, finite windows and a left-sided exponential

Find $X(z)$ and the ROC: (a) $x(n) = 2^n\sin(\omega_0 n)u(n)$ (b) $x(n) = \cos(\omega_0 n + \varphi)u(n)$ (c) $x(n) = (½)^n[u(n) - u(n-10)]$ (d) $x(n) = a^n u(k - n)$, $k > 0$.

1. **(a) Scaling.** Start from $\sin\omega_0 n\ u(n) \leftrightarrow \sin\omega_0\ z^{-1}/(1 - 2\cos\omega_0\ z^{-1} + z^{-2})$ and replace z^{-1} by $2z^{-1}$: $\mathbf{X(z) = 2\sin\omega_0\ z^{-1}/(1 - 4\cos\omega_0\ z^{-1} + 4z^{-2})}$. The poles are $2e^{\pm j\omega_0}$, radius 2, so **ROC $|z| > 2$**.
2. **(b) Expand the phase:** $\cos(\omega_0 n + \varphi) = \cos\varphi\cos\omega_0 n - \sin\varphi\sin\omega_0 n$. Combine the two table entries: numerator = $\cos\varphi(1 - \cos\omega_0 z^{-1}) - \sin\varphi\sin\omega_0 z^{-1} = \cos\varphi - z^{-1}(\cos\varphi\cos\omega_0 + \sin\varphi\sin\omega_0) = \cos\varphi - \cos(\omega_0 - \varphi)z^{-1}$.

$$X(z) = \frac{\cos\varphi - \cos(\omega_0 - \varphi)\ z^{-1}}{1 - 2\cos\omega_0\ z^{-1} + z^{-2}},\quad |z| > 1$$

1. **(c) Finite window.** Only $n = 0$ … 9 are nonzero, so $X(z) = \Sigma_{n=0}^{9}(½z^{-1})^n = \mathbf{(1 - (½z^{-1})^{10})/(1 - ½z^{-1})}$. The (apparent) pole at ½ is cancelled by a zero, so the only pole left is the 9-fold pole at $z = 0$. **ROC $|z| > 0$** (causal finite sequence).
2. **(d) Left-sided (right end at $n = k$).** $u(k - n) = 1$ for $n \le k$. Let $m = k - n$: $X(z) = \Sigma_{m=0}^{\infty}\ a^{k-m}z^{-(k-m)} = (a/z)^k\,\Sigma(z/a)^m = \mathbf{(a\ z^{-1})^k/(1 - a^{-1}z)}$, converging only when $\mathbf{|z| < |a|}$. For $k > 0$ the sequence is two-sided (it has samples at $n = 0$ … k), so $z = 0$ is also excluded because of the z^{-k} factor: **$0 < |z| < |a|$**. (If $k \le 0$ it is purely anti-causal and $|z| < |a|$ works with $z = 0$ included.)

Table rows 8–11 (slide 18) are the sinusoid entries. Whenever a phase φ appears, split with the cos/sin addition formula first, then combine over the common denominator.

L4-15 · Causal inverse with a mix of distinct and repeated poles

$X(z) = 1/[(1 + z^{-1})(1 - z^{-1})^2]$, causal. Find $x(n)$.

1. Write with unknown constants (one for each power of the repeated pole): $X(z) = A/(1 + z^{-1}) + B/(1 - z^{-1}) + C/(1 - z^{-1})^2$.
2. **A** (cover up $1 + z^{-1}$, set $z^{-1} = -1$): $A = 1/(1 + 1)^2 = ¼$. $\boldsymbol{C}$ (cover up $(1 - z^{-1})^2$, set $z^{-1} = 1$): $C = 1/(1 + 1) = ½$.
3. $\boldsymbol{B}$ from the value at $z^{-1} = 0$ (z → ∞): $1 = A + B + C \Rightarrow B = 1 - ¼ - ½ = ¼$.
4. Use $(1 - z^{-1})^{-2} \leftrightarrow (n + 1)u(n)$ (from $u * u$):

$$x(n) = [¼(-1)^n + ¼ + ½(n + 1)]\ u(n)$$

Check: $x(0) = ¼ + ¼ + ½ = 1$ ✓. The denominator $(1 + w)(1 - 2w + w^2) = 1 - w - w^2 + w^3$ gives $x(1) = x(0) = 1$, and the formula gives $-¼ + ¼ + 1 = 1$ ✓.

⚠ A double pole at p contributes $(A + Bn)p^n$. Here the repeated pole sits on the unit circle ($z = 1$), so $x(n)$ contains a term that *grows linearly* (½n): a causal system with this $H(z)$ would not be BIBO stable.

L4-16 · Convolution through Z, including a cancellation

Convolve with Z-transforms: (a) $x_1(n) = (⅓)^n u(n)$, $x_2(n) = (½)^n u(n)$ (b) $x_1(n) = \cos(\pi n/3)\,u(n)$, $x_2(n) = 0.5^n u(n)$.

1. **(a)** $Y(z) = 1/[(1 - ⅓z^{-1})(1 - ½z^{-1})]$, ROC $|z| > ½$. Residues: at the pole ⅓: $1/(1 - ½\cdot3) = \mathbf{-2}$; at the pole ½: $1/(1 - ⅓\cdot2) = \mathbf{3}$.
2. $\mathbf{y(n) = [3(½)^n - 2(⅓)^n]\,u(n)}$. Check $y(0) = 1$ ✓, $y(1) = 3/2 - 2/3 = 5/6 = ⅓ + ½$ ✓ (the convolution sum for $n = 1$).
3. **(b)** $X_1(z) = (1 - ½z^{-1})/(1 - z^{-1} + z^{-2})$ and $X_2(z) = 1/(1 - ½z^{-1})$. The numerator of X_1 **cancels** the pole of X_2:

$$Y(z) = X_1(z)X_2(z) = \frac{1}{1 - z^{-1} + z^{-2}},\quad |z| > 1$$

1. The poles of $1 - z^{-1} + z^{-2}$ are $e^{\pm j\pi/3}$ ($2r\cos\theta = 1$, $r = 1$). Using the sin ω_0 table row with $\omega_0 = \pi/3$: $\mathbf{y(n) = [\sin(\pi(n+1)/3)/\sin(\pi/3)]\,u(n) = (2/\sqrt{3})\sin(\pi(n+1)/3)\,u(n)}$.
2. Check: $y(0) = 1$, $y(1) = 1$ (= $\cos(\pi/3) + 0.5$ ✓), $y(2) = 0$, $y(3) = -1$.

A product can have a larger ROC than the overlap of the two ROCs when a pole of one is cancelled by a zero of the other. Here $|z| > ½$ and $|z| > 1$ overlap in $|z| > 1$, which still holds because the pole at ½ vanished.

L4-17 · System function, poles/zeros and $h(n)$ from a difference equation (improper H and unstable cases)

Find $H(z)$, the poles and zeros, and $h(n)$ (causal) for: (a) $y(n) = ½y(n-1) + 2x(n) + 0.5x(n-1) + x(n-2)$ (b) $y(n) - 3y(n-1) - 4y(n-2) = x(n) + 2x(n-1)$.

1. **(a)** Z-transform with zero initial conditions: $\mathbf{H(z) = (2 + 0.5z^{-1} + z^{-2})/(1 - ½z^{-1})}$ has a degree-2 numerator over a degree-1 denominator in z^{-1} ⇒ **improper** ($M \ge N$), so divide first.
2. With $w = z^{-1}$, long division gives $(w^2 + 0.5w + 2) \div (1 - 0.5w) = -2w - 5 + 7/(1 - 0.5w)$. So $\mathbf{H(z) = -5 - 2z^{-1} + 7/(1 - ½z^{-1})}$.
3. $\mathbf{h(n) = -5\,\delta(n) - 2\,\delta(n-1) + 7(½)^n\,u(n)}$. Check $h(0) = 2$ ✓, $h(1) = -2 + 3.5 = 1.5 = ½\cdot2 + 0.5$ ✓.
4. **Poles/zeros:** in positive powers $H = (2z^2 + 0.5z + 1)/(z(z - ½))$. **Poles** $z = 0$ and $z = ½$. **Zeros** from $2z^2 + 0.5z + 1 = 0$: $z = -0.125 \pm j0.696$ (complex pair, $|z| = 0.707$). Both poles are inside the unit circle ⇒ stable (ROC $|z| > ½$).
5. **(b)** $\mathbf{H(z) = (1 + 2z^{-1})/(1 - 3z^{-1} - 4z^{-2}) = (1 + 2z^{-1})/[(1 - 4z^{-1})(1 + z^{-1})]}$. In positive powers: $z(z + 2)/[(z - 4)(z + 1)]$.
6. **Zeros** $z = 0$ and $z = -2$. **Poles** $z = 4$ and $z = -1$. Residues: at 4: $(1 + ½)/(1 + ¼) = 1.2$; at −1: $(1 - 2)/(1 + 4) = -0.2$.

$$h(n) = [1.2(4)^n - 0.2(-1)^n]\ u(n)$$

1. Check $h(0) = 1$ ✓, $h(1) = 4.8 + 0.2 = 5 = 3\cdot1 + 2$ ✓. A pole at 4 is outside the unit circle (and −1 is on it) ⇒ the causal system is **unstable** (h grows like 4^n).

Improper $H(z)$ just means the system has a finite-length feedforward part added to the recursive part: the extra terms are delayed impulses.

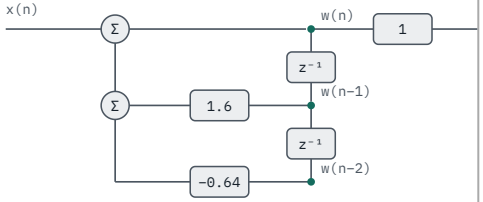
L4-18 · Cascade of two first-order sections: step response and Direct Form II

Two identical sections in cascade (each: adder, a z^{-1} delay, and a 0.8 feedback gain around the delay, so each section is $v(n) = u(n) + 0.8\,v(n-1)$). The input $x(n)$ enters the first section, the output of the first feeds the second, and $y(n)$ is the output of the second. (a) Find $H(z)$. (b) Find the unit-step response (initially relaxed). (c) Draw the Direct Form II equivalent.

1. **(a)** Each section: $H_1(z) = 1/(1 - 0.8z^{-1})$. Cascade ⇒ product: $\mathbf{H(z) = 1/(1 - 0.8z^{-1})^2 = 1/(1 - 1.6z^{-1} + 0.64z^{-2})}$ (a double pole at 0.8, so stable and causal).
2. **(b)** $Y(z) = H(z)/(1 - z^{-1}) = A/(1 - z^{-1}) + B/(1 - 0.8z^{-1}) + C/(1 - 0.8z^{-1})^2$. **A** = $1/(1 - 0.8)^2 = \mathbf{25}$. $\boldsymbol{C}$ (at $z^{-1} = 1.25$) = $1/(1 - 1.25) = \mathbf{-4}$. $\boldsymbol{B}$ from $y(0) = 1 \Rightarrow 1 = 25 + B - 4 \Rightarrow \mathbf{B = -20}$.
3. Use $(1 - pz^{-1})^{-2} \leftrightarrow (n + 1)p^n\,u(n)$:

$$y(n) = [25 - 20(0.8)^n - 4(n + 1)(0.8)^n]\ u(n)$$

1. Check: $y(0) = 25 - 20 - 4 = 1$ ✓; $y(1) = 25 - 16 - 6.4 = 2.6 = 1.6\cdot1 + 1$ ✓. As $n \to \infty$, $y \to 25 = H(1) = 1/(1 - 0.8)^2$ ✓ (DC gain).
2. **(c) Direct Form II:** multiply out the denominator; the difference equation is $y(n) = 1.6\,y(n-1) - 0.64\,y(n-2) + x(n)$. Since there are no zeros ($b_0 = 1$ only), DF-I and DF-II coincide: one shared delay line with feedback taps +1.6 and −0.64.



Direct Form II of $1/(1 - 1.6z^{-1} + 0.64z^{-2})$: drawn feedback gains +1.6 on $w(n-1)$ and −0.64 on $w(n-2)$. Two delays, the same count as the cascade.

The cascade and the single second-order section use the same number of delays (2), but the cascade has two separate gain-0.8 loops. Multiplying the section polynomials $(1 - 0.8z^{-1})^2$ gives the DF-II coefficients $1.6 = 2\cdot0.8$ and $0.64 = 0.8^2$.